Policy-match donor counties as placebos for Multnomah

Page	County	Pop. 2020	Used	(a) Pre-trend p	(b) IRS dist.	Out-migration	Out-of-state	In-migration
3	**Multnomah, Oregon**	**815,428**	**treated**	**0.39**		**1.88 (0.57)**	**1.34 (0.44)**	**-0.0028 (0.0020)**
4	Orleans Parish, Louisiana	383,997	yes	0.28	0.31	-2.10 (0.91)	-2.18 (0.78)	-0.0020 (0.0009)
5	Dakota, Minnesota	439,882	yes	0.30	0.31	1.33 (0.80)	0.48 (0.43)	0.0028 (0.0022)
6	Hennepin, Minnesota	1,281,565	yes	0.97	0.24	0.43 (0.51)	0.25 (0.29)	-0.0032 (0.0030)
7	Camden, New Jersey	523,485	yes	0.62	0.41	-0.83 (0.70)	-1.31 (0.55)	0.0026 (0.0017)
8	Union, New Jersey	575,345	yes	0.14	0.35	-0.00 (0.62)	0.04 (0.42)	-0.0011 (0.0015)
9	Monroe, New York	759,443	yes	0.13	0.42	0.57 (0.43)	0.24 (0.36)	0.0011 (0.0010)
10	Providence, Rhode Island	660,741	yes	0.12	0.43	0.03 (0.59)	0.22 (0.51)	0.0004 (0.0009)
11	Virginia Beach city, Virginia	459,470	yes	0.21	0.42	-0.08 (0.76)	0.24 (0.58)	-0.0008 (0.0013)
12	New Castle, Delaware	570,719	no	0.14	0.48	0.60 (0.60)	0.85 (0.56)	0.0000 (0.0014)
13	Jefferson Parish, Louisiana	440,781	no	n/a	0.22	n/a	n/a	n/a
14	Anoka, Minnesota	363,887	no	0.07	0.30	0.34 (0.83)	0.14 (0.31)	0.0001 (0.0007)
15	Burlington, New Jersey	461,860	no	0.39	0.52	-0.35 (0.63)	-0.80 (0.48)	0.0013 (0.0009)
16	Essex, New Jersey	863,728	no	0.55	0.89	-0.64 (0.52)	-0.23 (0.37)	0.0020 (0.0017)
17	Gloucester, New Jersey	302,294	no	0.02	0.51	1.08 (0.76)	0.91 (0.52)	0.0004 (0.0006)
18	Hudson, New Jersey	724,854	no	0.44	0.58	0.35 (0.68)	0.01 (0.54)	0.0000 (0.0015)
19	Middlesex, New Jersey	863,162	no	0.45	0.47	0.24 (0.50)	0.02 (0.33)	-0.0013 (0.0019)
20	Erie, New York	954,236	no	0.43	0.44	-0.52 (0.38)	-0.39 (0.33)	-0.0001 (0.0011)
21	Richmond, New York	495,747	no	0.42	0.50	1.02 (0.55)	0.70 (0.49)	0.0004 (0.0006)
22	Henrico, Virginia	334,389	no	0.97	0.63	-0.52 (0.84)	-0.45 (0.48)	0.0011 (0.0010)
23	Richmond city, Virginia	226,610	no	0.19	0.60	0.54 (1.30)	-0.34 (0.64)	-0.0000 (0.0010)

How to read these pages. Each Policy-match donor county is treated as if it were Multnomah in the individual-level difference-in-differences (College × Post, ACS, 2016–2024 without 2020). A donor is used as a placebo only if it passes two pre-2020 screens: (a) its own out-migration event study shows no college / non-college pre-trend (joint p-value on the 2016–2018 College × Year coefficients above the threshold in globals.do); (b) its 2016–2019 IRS AGI out- and in-migration rate slopes are among the closest half of donors to Multnomah's (distance at most 0.43). The three right-hand columns are the College × Post estimates in percentage points, standard errors in parentheses.
IRS windows on the county pages: 2018–19 (before) and 2021–22 (after), by filing year; 2020 is left out.

Migration concentration before and after 2020

County	Out: border cty	Out: in-state	Out: border st.	Out: top 10	In: border cty	In: in-state	In: border st.	In: top 10
Multnomah, Oregon	**46 → 43**	**47 → 45**	**27 → 27**	**56 → 54**	**35 → 34**	**39 → 39**	**28 → 30**	**48 → 48**
Donor median	**43 → 40**	**55 → 54**	**11 → 12**	**56 → 52**	**43 → 44**	**57 → 57**	**10 → 9**	**58 → 59**
Orleans Parish, Louisiana	40 → 39	49 → 49	10 → 12	51 → 52	35 → 36	47 → 50	8 → 8	46 → 48
Dakota, Minnesota	59 → 55	71 → 69	4 → 4	64 → 61	61 → 63	73 → 76	3 → 3	66 → 68
Hennepin, Minnesota	49 → 47	61 → 60	5 → 5	55 → 54	44 → 46	57 → 60	8 → 8	50 → 52
Camden, New Jersey	54 → 51	55 → 54	19 → 18	62 → 60	57 → 55	57 → 53	24 → 28	66 → 65
Union, New Jersey	47 → 45	62 → 61	11 → 11	60 → 57	55 → 52	64 → 61	19 → 23	70 → 70
Monroe, New York	22 → 22	41 → 41	6 → 6	32 → 31	26 → 23	53 → 52	6 → 6	38 → 35
Providence, Rhode Island	41 → 36	29 → 28	33 → 29	53 → 49	41 → 38	26 → 24	42 → 43	57 → 55
Virginia Beach city, Virginia	32 → 32	47 → 50	7 → 8	43 → 45	31 → 31	46 → 47	6 → 6	42 → 42
New Castle, Delaware	30 → 30	15 → 16	43 → 40	45 → 44	32 → 30	10 → 10	52 → 52	47 → 48
Jefferson Parish, Louisiana	50 → 47	62 → 60	10 → 12	59 → 57	54 → 56	67 → 68	7 → 6	62 → 64
Anoka, Minnesota	63 → 59	80 → 78	2 → 2	71 → 67	68 → 69	83 → 84	2 → 1	75 → 77
Burlington, New Jersey	43 → 40	45 → 45	17 → 16	50 → 48	50 → 49	56 → 56	17 → 20	59 → 59
Essex, New Jersey	39 → 37	55 → 55	17 → 15	55 → 52	43 → 41	53 → 51	28 → 30	68 → 68
Gloucester, New Jersey	57 → 54	58 → 57	16 → 15	66 → 64	65 → 61	66 → 62	17 → 21	74 → 72
Hudson, New Jersey	39 → 34	47 → 48	23 → 20	56 → 52	41 → 42	35 → 33	39 → 42	58 → 59
Middlesex, New Jersey	33 → 32	54 → 54	13 → 13	50 → 49	37 → 36	57 → 58	19 → 22	58 → 61
Erie, New York	26 → 25	43 → 42	4 → 4	35 → 34	29 → 25	52 → 54	4 → 4	42 → 41
Richmond, New York	43 → 40	37 → 30	35 → 37	57 → 51	64 → 68	73 → 75	11 → 10	77 → 79
Henrico, Virginia	53 → 53	71 → 72	4 → 3	57 → 57	50 → 49	70 → 71	4 → 4	54 → 54
Richmond city, Virginia	51 → 49	73 → 73	4 → 5	61 → 59	42 → 37	71 → 67	5 → 6	54 → 50

Each cell is the 2018–19 share → the 2021–22 share, in percent of the county's domestic movers (IRS returns, summed over the window's filing years). "Out" columns describe where people leaving the county went; "In" columns where arrivals came from.

- Border county: a county adjacent to it (Census county adjacency file, water boundaries included), in any state.
- In-state: elsewhere in its own state. Exact, from the IRS same-state and different-state totals.
- Border state: a state that shares a border with its own. This is a lower bound: the IRS reports a county pair only above a disclosure threshold, so small flows to scattered counties in those states are not counted.
- Top 10: its 10 largest county-to-county flows in that window.

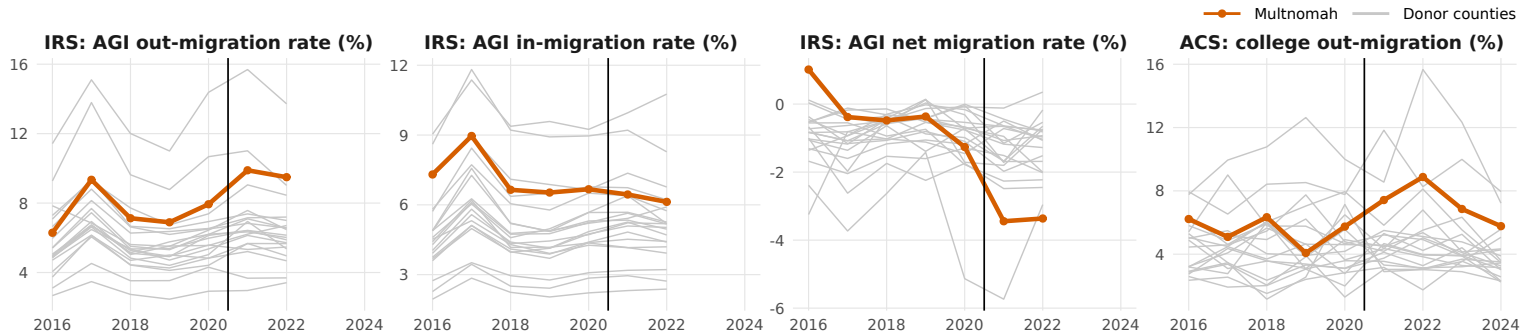
The same measures on an AGI basis are in flow_concentration.csv, with the share of movers the IRS discloses by county pair.

Multnomah County, Oregon

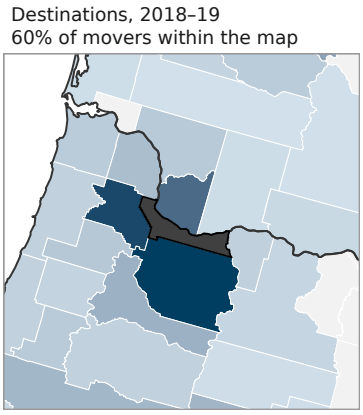
Treated county (Preschool for All income tax, effective 2021)

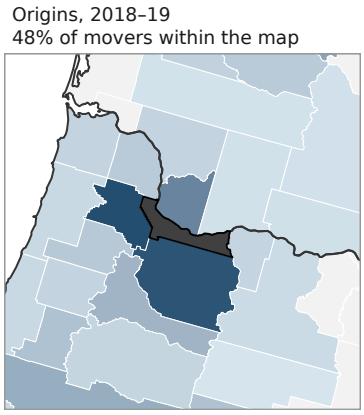
→ = 2018–19 to 2021–22 average	Multnomah	College × Post DiD (the treated estimates)	Multnomah
Population, 2020 Census	815,428	Own pre-trend, joint p-value (screen a)	0.39
Density (people per sq. mile)	1,891	AGI out / in slope, 2016–19 (pp/yr)	-0.04 / -0.46
Urban share of population	99%	Donor IRS-slope distance cutoff (screen b)	0.43
Median household income (ACS)	$69,176	Donors passing both screens	8 of 20
Share under 24 / 65 and over	27% / 13%	DiD, out-migration: pp (se)	1.88 (0.57)
State top income-tax rate, 2019 / 2020 (%)	9.90 / 9.90	DiD, out-of-state: pp (se)	1.34 (0.44)
Governor's party, 2019 / 2020	D / D	DiD, in-migration: pp (se)	-0.0028 (0.0020)
Days under stay-at-home order	88		
IRS AGI out-migration rate (%)	7.01 → 9.69		
IRS AGI in-migration rate (%)	6.59 → 6.29		
IRS AGI net migration rate (%)	-0.42 → -3.40		

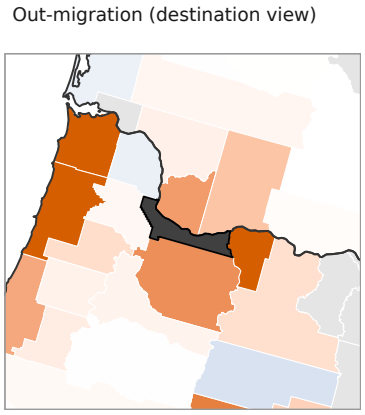
Migration rates before and after 2020



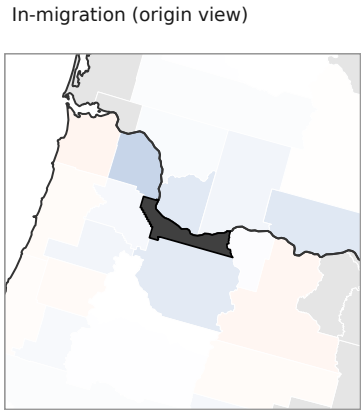
Where its movers went and came from

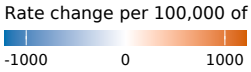






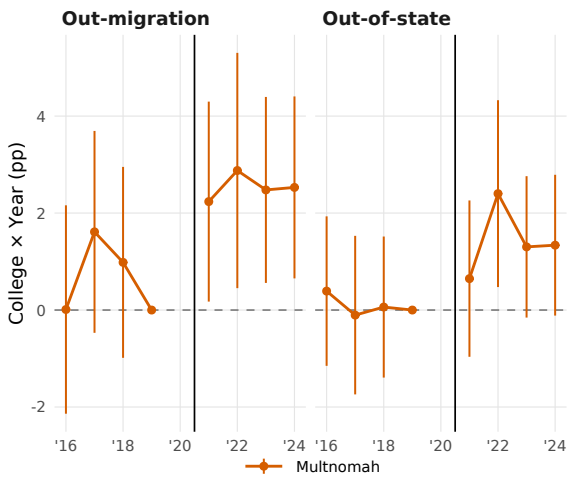
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22
Top destinations			**Out-movers, to:**		
Clackamas, OR	18.3	17.3	Border county	46	43
Washington, OR	16.1	14.6	Within state	47	45
Clark, WA	9.6	9.7	Bordering state	27	27
King, WA	3.0	2.6	Top 10 flows	56	54
Los Angeles, CA	2.1	2.0	**In-movers, from:**		
Top origins			Border county	35	34
Washington, OR	14.7	14.3	Within state	39	39
Clackamas, OR	13.6	12.9	Bordering state	28	30
Clark, WA	6.0	5.8	Top 10 flows	48	48
King, WA	3.0	3.1			
Los Angeles, CA	3.0	3.6			

DiD event study (base 2019, 95% CI)



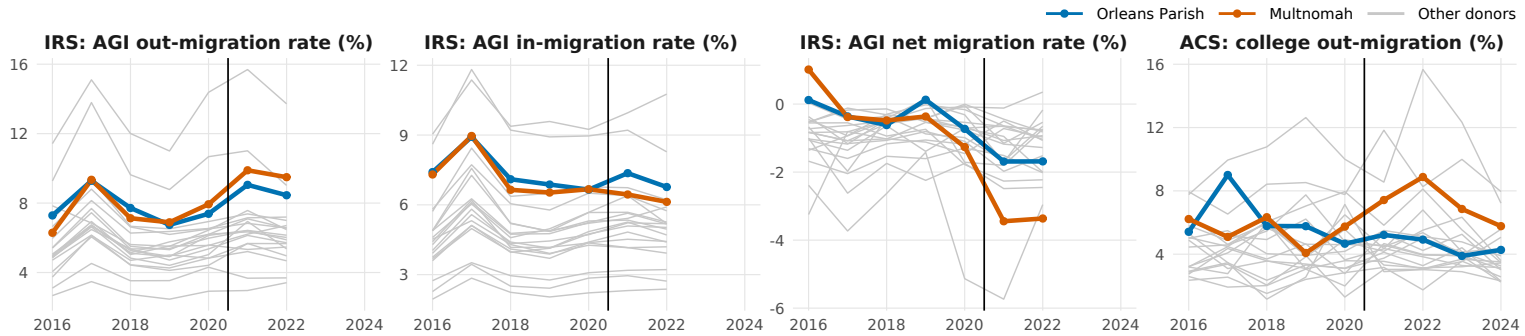
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Orleans Parish, Louisiana

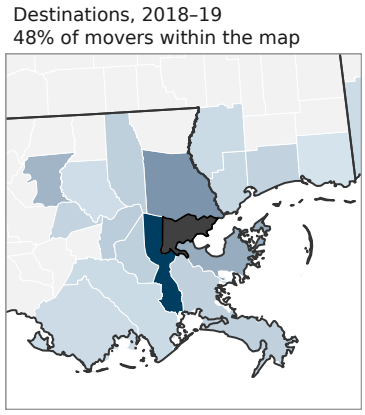
Used as a placebo: passes both pre-2020 screens

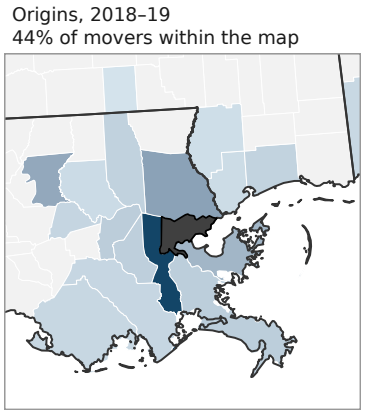
→ = 2018–19 to 2021–22 average	Orleans Parish	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	383,997	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	2,266	1,891	Origin county identified in the ACS?	yes	
Urban share of population	100%	99%	(a) Own pre-trend, joint p-value	0.28 — pass	0.39
Median household income (ACS)	$41,604	$69,176	(b) IRS slope distance from Multnomah	0.31 — pass	cutoff 0.43
Share under 24 / 65 and over	29% / 14%	27% / 13%	rank among donors (1 = closest)	4 of 20	
State top income-tax rate, 2019 / 2020 (%)	6.00 / 6.00	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.32 / -0.34	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	53	88	Other donor pools	none	
IRS AGI out-migration rate (%)	7.23 → 8.75	7.01 → 9.69	DiD, out-migration: pp (se)	-2.10 (0.91)	1.88 (0.57)
IRS AGI in-migration rate (%)	6.99 → 7.06	6.59 → 6.29	DiD, out-of-state: pp (se)	-2.18 (0.78)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.24 → -1.69	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0020 (0.0009)	-0.0028 (0.0020)

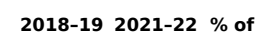
Migration rates before and after 2020



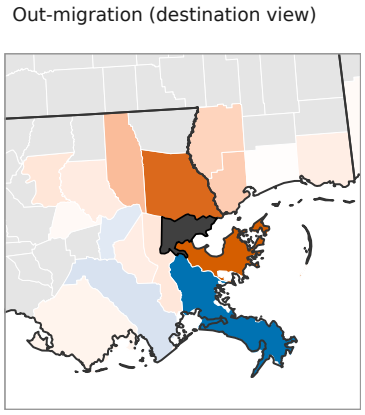
Where its movers went and came from

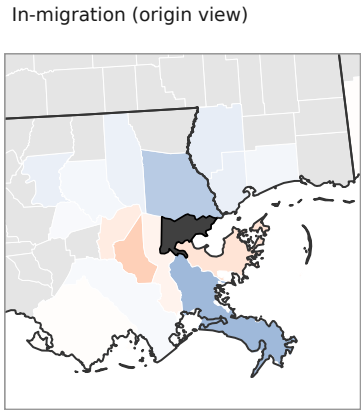






Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)

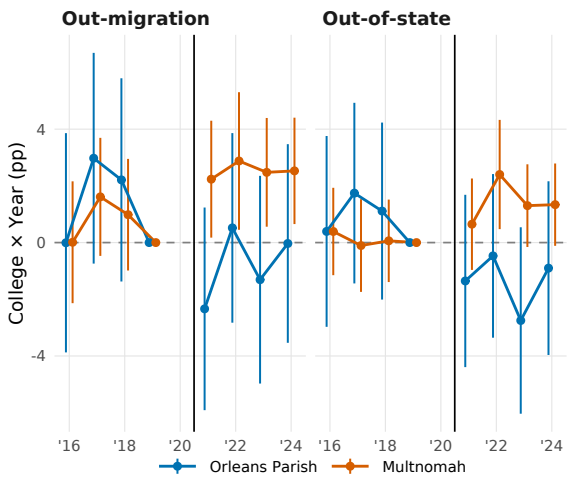






% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Jefferson, LA	29.2	25.9	Border county	40	39	46	43
St. Tammany, LA	6.6	8.6	Within state	49	49	47	45
Harris, TX	3.7	4.9	Bordering state	10	12	27	27
St. Bernard, LA	3.5	3.9	Top 10 flows	51	52	56	54
East Baton Rouge, LA	2.6	2.7	**In-movers, from:**				
Top origins			Border county	35	36	35	34
Jefferson, LA	26.6	27.6	Within state	47	50	39	39
St. Tammany, LA	4.7	4.9	Bordering state	8	8	28	30
East Baton Rouge, LA	4.0	4.0	Top 10 flows	46	48	48	48
Harris, TX	3.0	2.7					
St. Bernard, LA	2.5	2.7					

Placebo event study (base 2019, 95% CI)



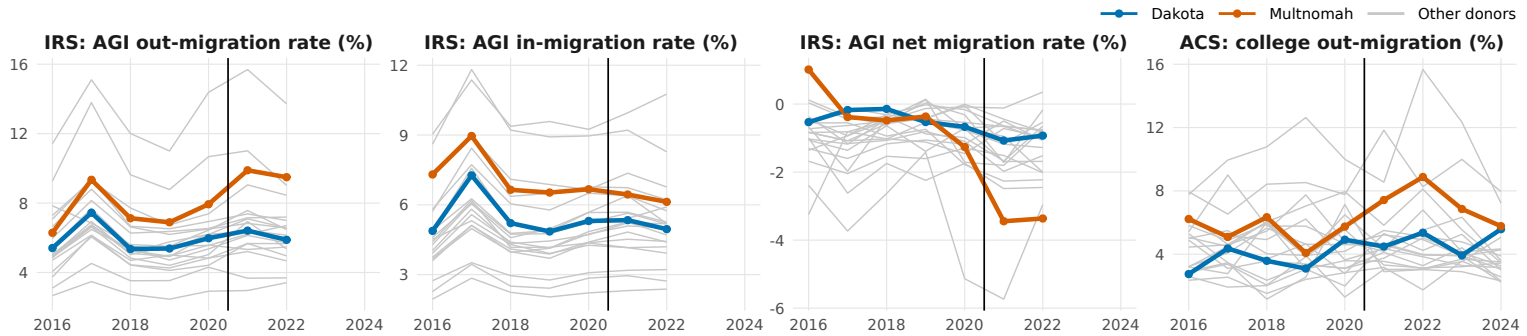
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Dakota County, Minnesota

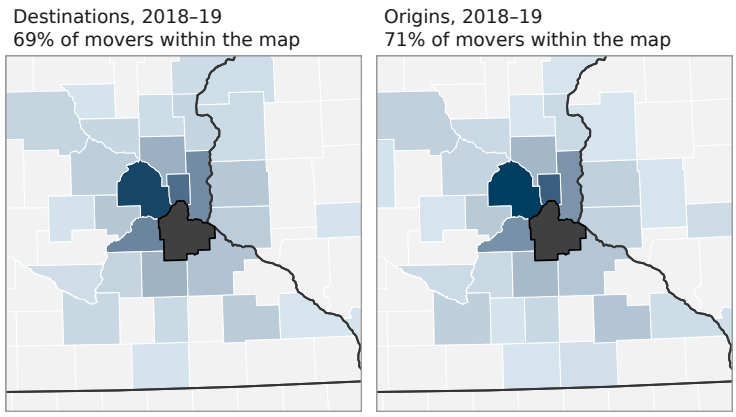
Used as a placebo: passes both pre-2020 screens

→ = 2018–19 to 2021–22 average	Dakota	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	439,882	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	782	1,891	Origin county identified in the ACS?	yes	
Urban share of population	95%	99%	(a) Own pre-trend, joint p-value	0.30 — pass	0.39
Median household income (ACS)	$86,036	$69,176	(b) IRS slope distance from Multnomah	0.31 — pass	cutoff 0.43
Share under 24 / 65 and over	33% / 14%	27% / 13%	rank among donors (1 = closest)	5 of 20	
State top income-tax rate, 2019 / 2020 (%)	9.85 / 9.85	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.22 / -0.21	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	51	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	5.37 → 6.15	7.01 → 9.69	DiD, out-migration: pp (se)	1.33 (0.80)	1.88 (0.57)
IRS AGI in-migration rate (%)	5.03 → 5.15	6.59 → 6.29	DiD, out-of-state: pp (se)	0.48 (0.43)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.34 → -1.00	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0028 (0.0022)	-0.0028 (0.0020)

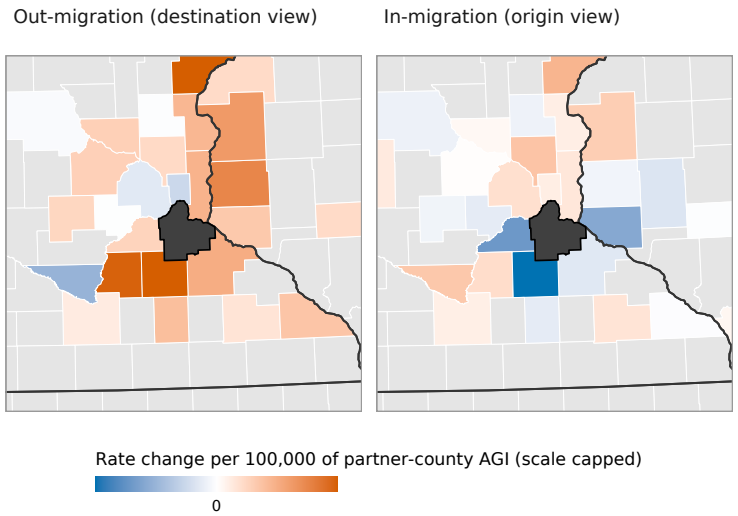
Migration rates before and after 2020



Where its movers went and came from

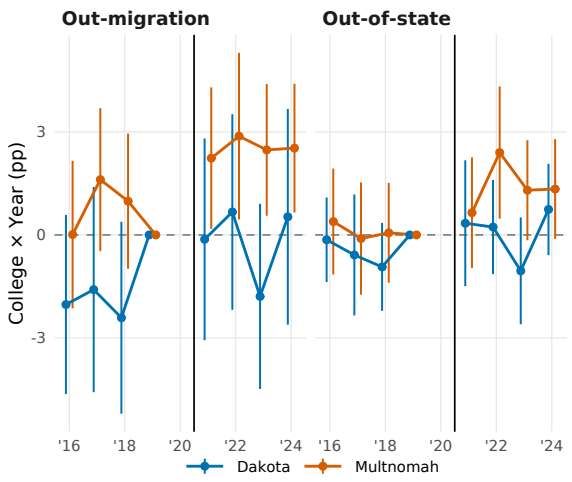


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Hennepin, MN	24.4	22.0	Border county	59	55	46	43
Ramsey, MN	13.4	12.1	Within state	71	69	47	45
Scott, MN	8.6	8.1	Bordering state	4	4	27	27
Washington, MN	7.5	7.5	Top 10 flows	64	61	56	54
Anoka, MN	2.8	3.0	**In-movers, from:**				
Top origins			Border county	61	63	35	34
Hennepin, MN	27.0	28.8	Within state	73	76	39	39
Ramsey, MN	17.3	17.5	Bordering state	3	3	28	30
Scott, MN	6.8	6.7	Top 10 flows	66	68	48	48
Washington, MN	6.3	6.6					
Anoka, MN	2.3	2.6					

Placebo event study (base 2019, 95% CI)



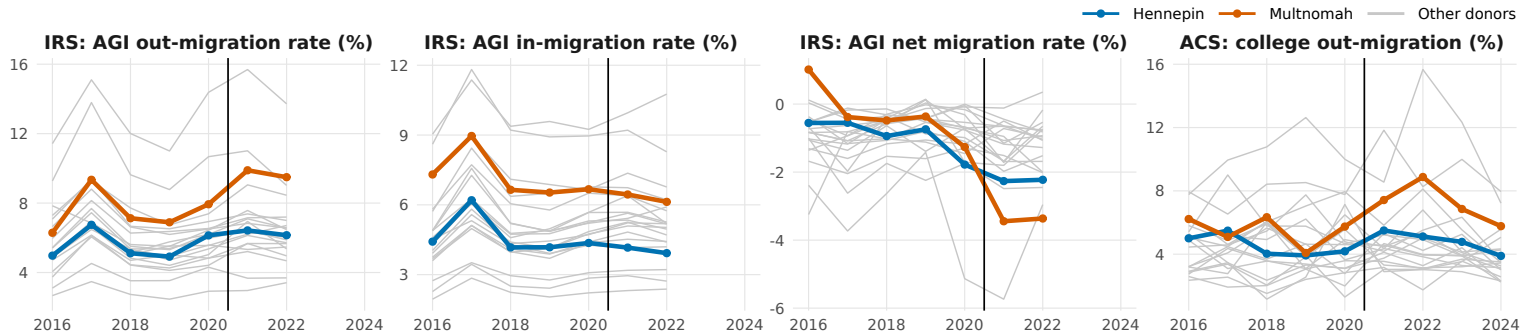
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Hennepin County, Minnesota

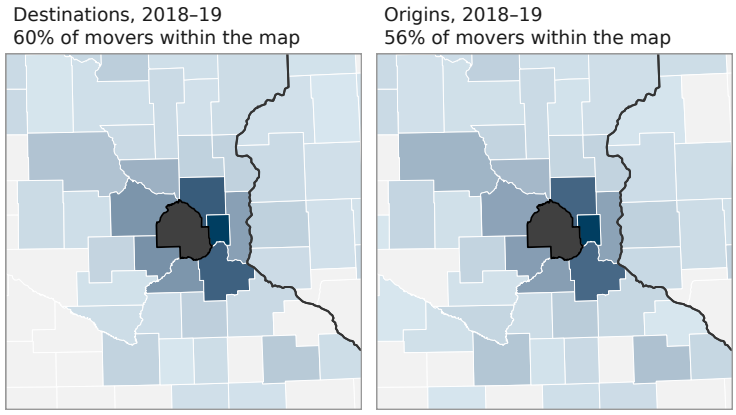
Used as a placebo: passes both pre-2020 screens

→ = 2018–19 to 2021–22 average	Hennepin	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	1,281,565	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	2,313	1,891	Origin county identified in the ACS?	yes	
Urban share of population	98%	99%	(a) Own pre-trend, joint p-value	0.97 — pass	0.39
Median household income (ACS)	$78,167	$69,176	(b) IRS slope distance from Multnomah	0.24 — pass	cutoff 0.43
Share under 24 / 65 and over	31% / 14%	27% / 13%	rank among donors (1 = closest)	2 of 20	
State top income-tax rate, 2019 / 2020 (%)	9.85 / 9.85	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.18 / -0.28	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	51	88	Other donor pools	stringency; narrow; demog.	
IRS AGI out-migration rate (%)	5.01 → 6.29	7.01 → 9.69	DiD, out-migration: pp (se)	0.43 (0.51)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.17 → 4.04	6.59 → 6.29	DiD, out-of-state: pp (se)	0.25 (0.29)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.84 → -2.25	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0032 (0.0030)	-0.0028 (0.0020)

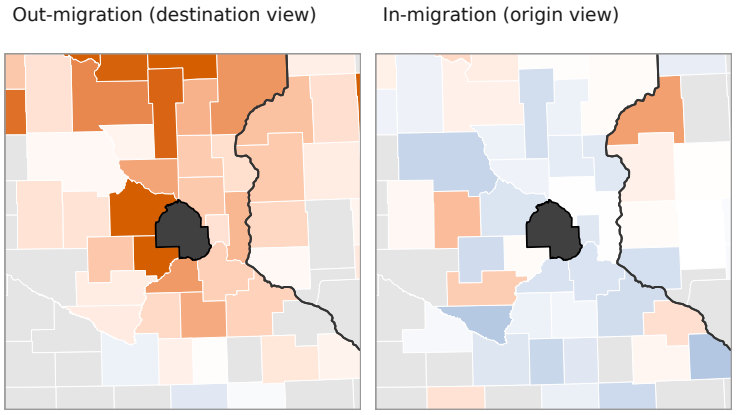
Migration rates before and after 2020



Where its movers went and came from

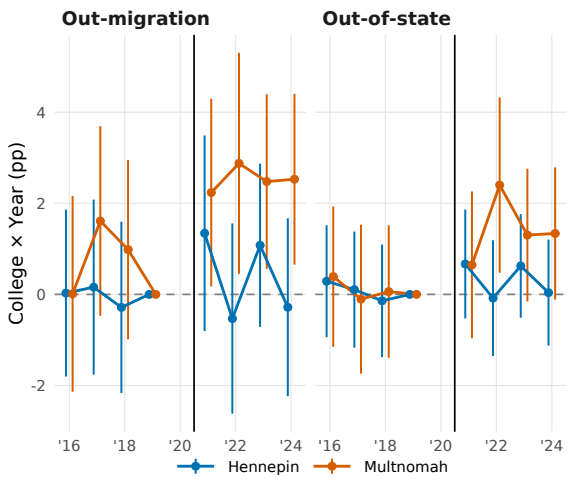


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Ramsey, MN	16.1	14.3	Border county	49	47	46	43
Anoka, MN	10.6	10.3	Within state	61	60	47	45
Dakota, MN	9.8	9.3	Bordering state	5	5	27	27
Carver, MN	4.0	3.9	Top 10 flows	55	54	56	54
Wright, MN	3.7	4.1	**In-movers, from:**				
Top origins			Border county	44	46	35	34
Ramsey, MN	16.0	17.0	Within state	57	60	39	39
Anoka, MN	9.1	9.5	Bordering state	8	8	28	30
Dakota, MN	8.7	8.8	Top 10 flows	50	52	48	48
Carver, MN	3.0	3.2					
Scott, MN	2.9	3.1					

Placebo event study (base 2019, 95% CI)



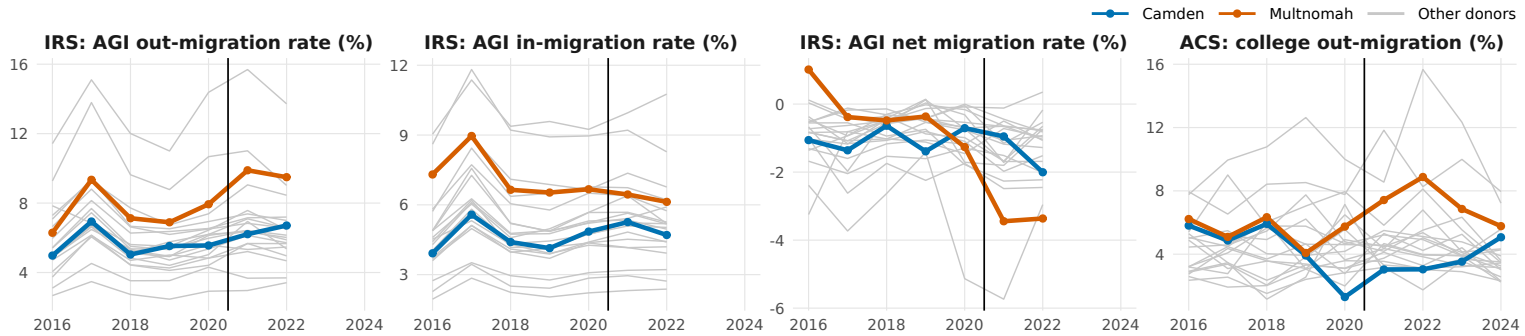
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Camden County, New Jersey

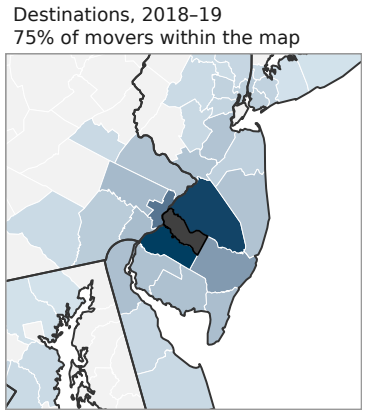
Used as a placebo: passes both pre-2020 screens

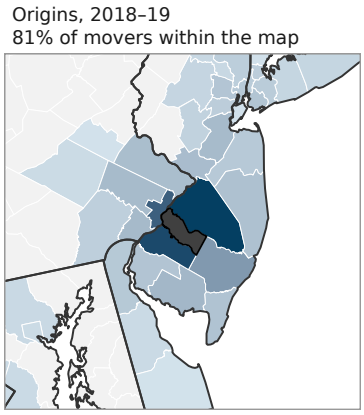
→ = 2018–19 to 2021–22 average	Camden	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	523,485	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	2,365	1,891	Origin county identified in the ACS?	yes	
Urban share of population	99%	99%	(a) Own pre-trend, joint p-value	0.62 — pass	0.39
Median household income (ACS)	$70,451	$69,176	(b) IRS slope distance from Multnomah	0.41 — pass	cutoff 0.43
Share under 24 / 65 and over	31% / 15%	27% / 13%	rank among donors (1 = closest)	7 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.02 / -0.05	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	5.28 → 6.46	7.01 → 9.69	DiD, out-migration: pp (se)	-0.83 (0.70)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.27 → 4.98	6.59 → 6.29	DiD, out-of-state: pp (se)	-1.31 (0.55)	1.34 (0.44)
IRS AGI net migration rate (%)	-1.01 → -1.48	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0026 (0.0017)	-0.0028 (0.0020)

Migration rates before and after 2020

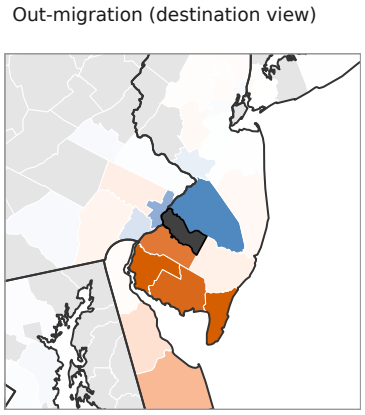


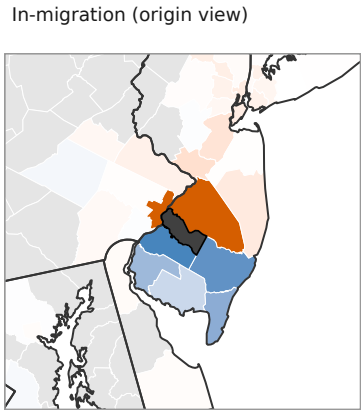
Where its movers went and came from





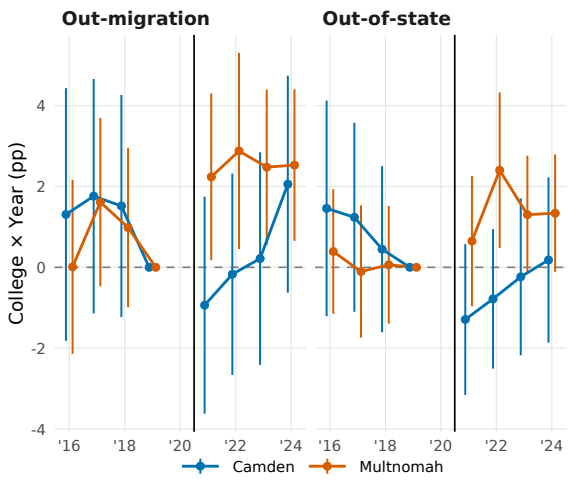
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Gloucester, NJ	21.1	20.1	Border county	54	51	46	43
Burlington, NJ	19.5	18.4	Within state	55	54	47	45
Philadelphia, PA	9.2	8.3	Bordering state	19	18	27	27
Atlantic, NJ	4.3	4.3	Top 10 flows	62	60	56	54
Montgomery, PA	1.6	1.6	**In-movers, from:**				
Top origins			Border county	57	55	35	34
Burlington, NJ	20.7	18.6	Within state	57	53	39	39
Gloucester, NJ	18.3	16.0	Bordering state	24	28	28	30
Philadelphia, PA	13.2	16.3	Top 10 flows	66	65	48	48
Atlantic, NJ	4.4	4.0					
Cumberland, NJ	1.8	1.7					

Placebo event study (base 2019, 95% CI)



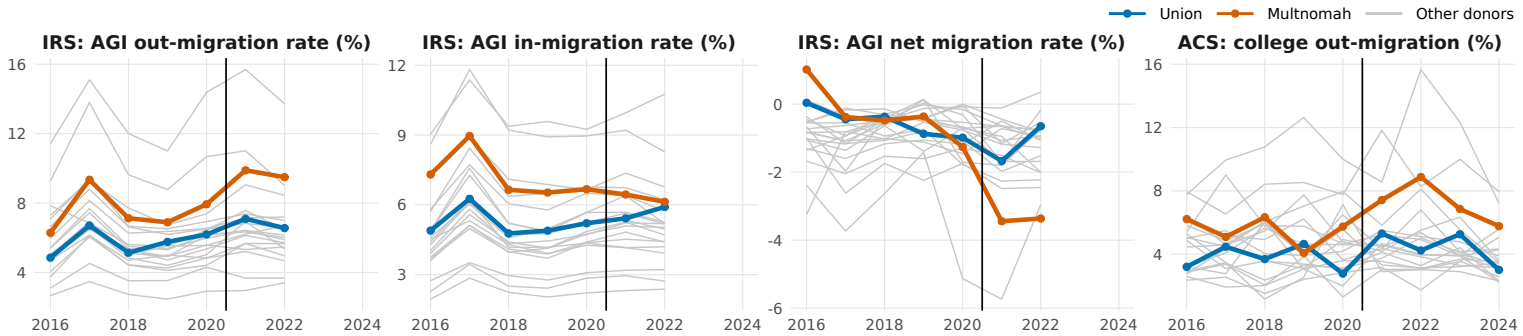
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Union County, New Jersey

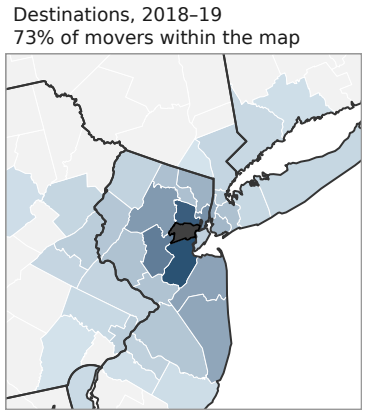
Used as a placebo: passes both pre-2020 screens

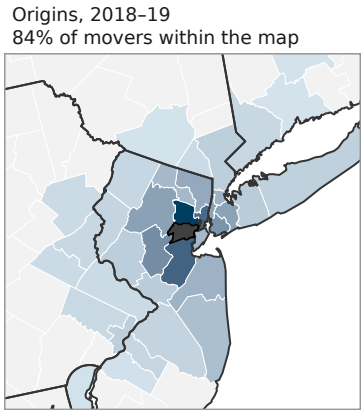
→ = 2018–19 to 2021–22 average	Union	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	575,345	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	5,599	1,891	Origin county identified in the ACS?	yes	
Urban share of population	100%	99%	(a) Own pre-trend, joint p-value	0.14 — pass	0.39
Median household income (ACS)	$80,198	$69,176	(b) IRS slope distance from Multnomah	0.35 — pass	cutoff 0.43
Share under 24 / 65 and over	32% / 14%	27% / 13%	rank among donors (1 = closest)	6 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	0.12 / -0.15	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	5.45 → 6.83	7.01 → 9.69	DiD, out-migration: pp (se)	-0.00 (0.62)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.83 → 5.66	6.59 → 6.29	DiD, out-of-state: pp (se)	0.04 (0.42)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.62 → -1.16	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0011 (0.0015)	-0.0028 (0.0020)

Migration rates before and after 2020

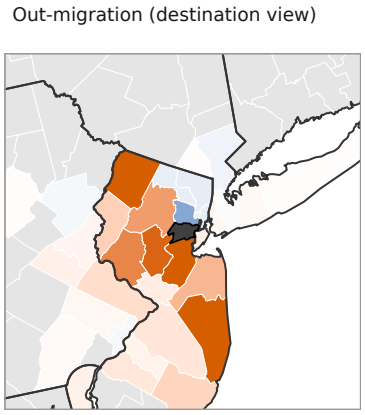


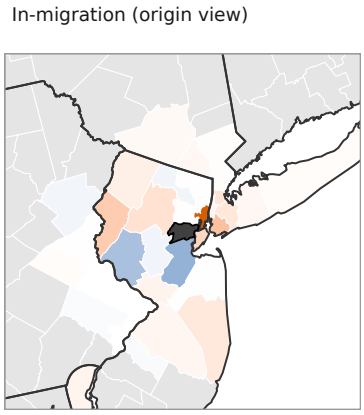
Where its movers went and came from





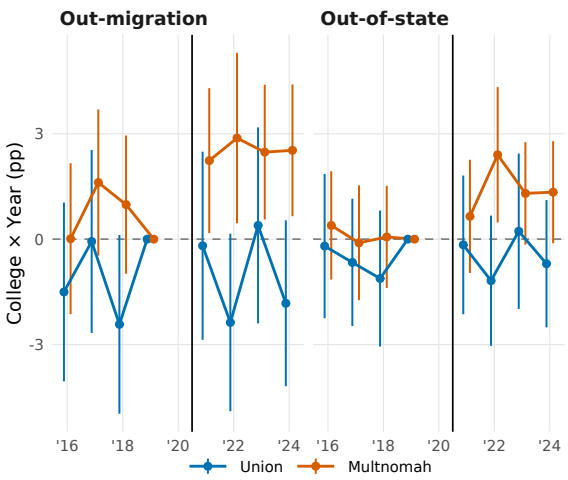
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Middlesex, NJ	16.0	16.1	Border county	47	45	46	43
Essex, NJ	13.5	10.8	Within state	62	61	47	45
Somerset, NJ	7.9	7.9	Bordering state	11	11	27	27
Hudson, NJ	5.3	5.2	Top 10 flows	60	57	56	54
Morris, NJ	4.2	4.4	**In-movers, from:**				
Top origins			Border county	55	52	35	34
Essex, NJ	21.0	19.4	Within state	64	61	39	39
Middlesex, NJ	12.3	10.5	Bordering state	19	23	28	30
Hudson, NJ	11.1	12.1	Top 10 flows	70	70	48	48
Somerset, NJ	5.7	5.1					
Kings, NY	4.9	7.0					

Placebo event study (base 2019, 95% CI)



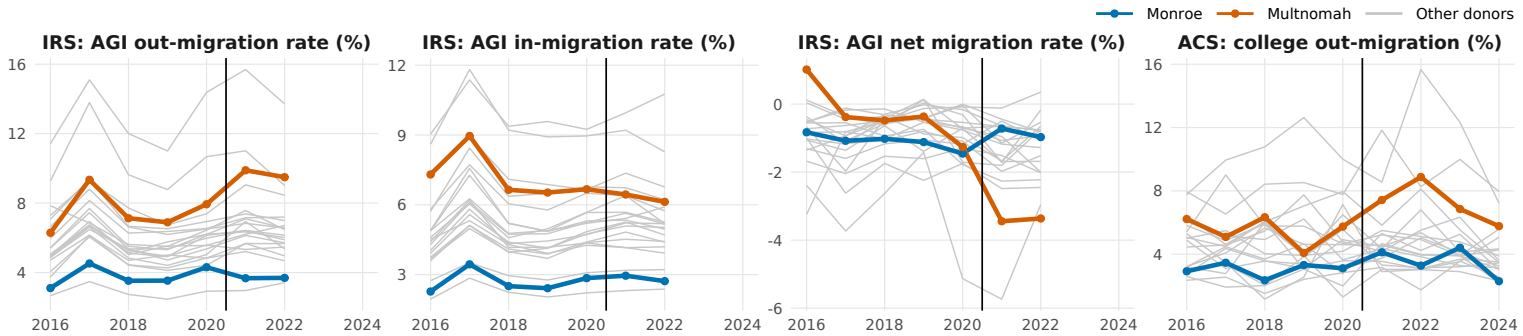
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Monroe County, New York

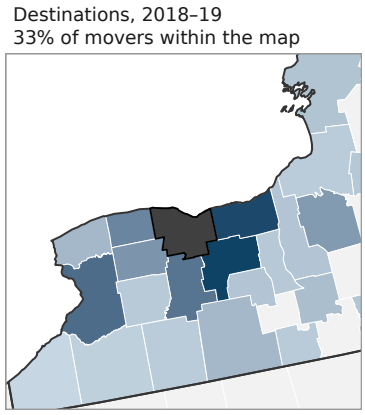
Used as a placebo: passes both pre-2020 screens

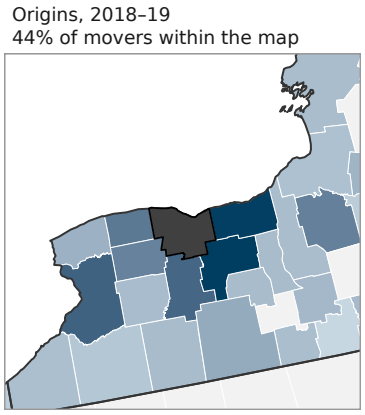
→ = 2018–19 to 2021–22 average	Monroe	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	759,443	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	1,156	1,891	Origin county identified in the ACS?	yes	
Urban share of population	92%	99%	(a) Own pre-trend, joint p-value	0.13 — pass	0.39
Median household income (ACS)	$60,075	$69,176	(b) IRS slope distance from Multnomah	0.42 — pass	cutoff 0.43
Share under 24 / 65 and over	31% / 17%	27% / 13%	rank among donors (1 = closest)	8 of 20	
State top income-tax rate, 2019 / 2020 (%)	8.82 / 8.82	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	0.03 / -0.05	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	285	88	Other donor pools	urban-COVID; demog.	
IRS AGI out-migration rate (%)	3.53 → 3.68	7.01 → 9.69	DiD, out-migration: pp (se)	0.57 (0.43)	1.88 (0.57)
IRS AGI in-migration rate (%)	2.46 → 2.84	6.59 → 6.29	DiD, out-of-state: pp (se)	0.24 (0.36)	1.34 (0.44)
IRS AGI net migration rate (%)	-1.07 → -0.84	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0011 (0.0010)	-0.0028 (0.0020)

Migration rates before and after 2020

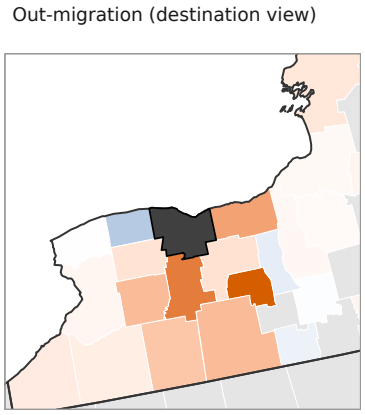


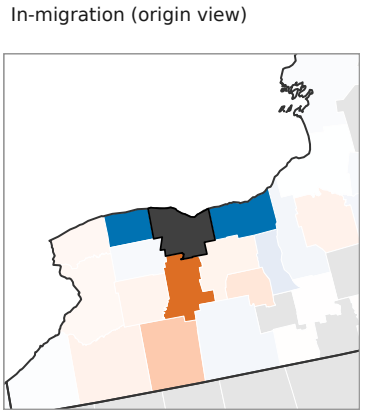
Where its movers went and came from





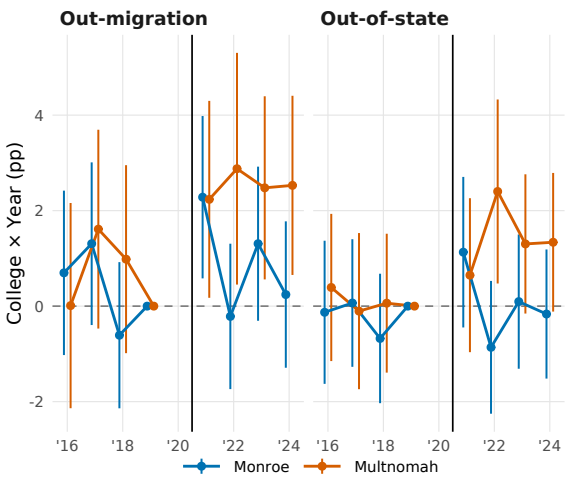
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Ontario, NY	7.5	7.3	Border county	22	22	46	43
Wayne, NY	6.9	6.7	Within state	41	41	47	45
Erie, NY	4.0	4.0	Bordering state	6	6	27	27
Livingston, NY	3.5	3.5	Top 10 flows	32	31	56	54
Orleans, NY	2.6	2.3	**In-movers, from:**				
Top origins			Border county	26	23	35	34
Ontario, NY	8.0	7.1	Within state	53	52	39	39
Wayne, NY	7.9	6.4	Bordering state	6	6	28	30
Erie, NY	4.8	5.1	Top 10 flows	38	35	48	48
Livingston, NY	4.5	4.2					
Orleans, NY	3.3	2.7					

Placebo event study (base 2019, 95% CI)



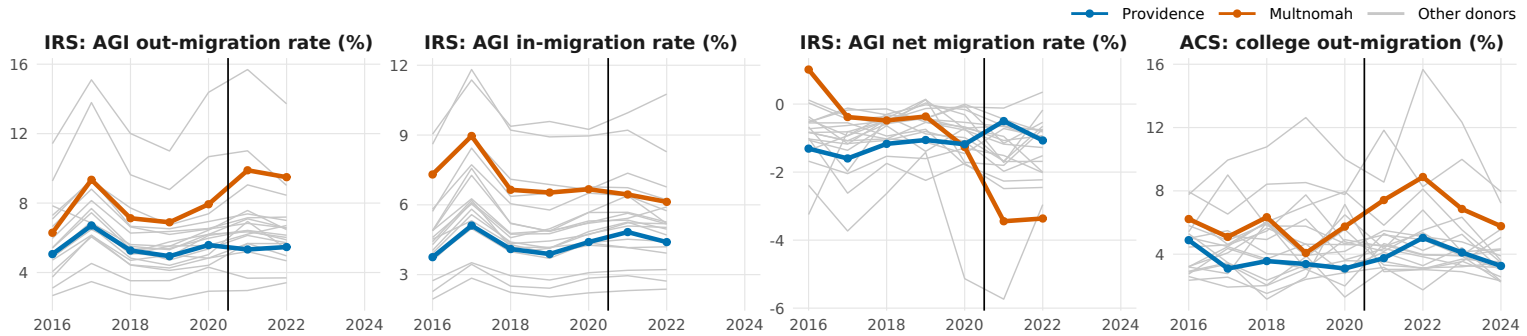
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Providence County, Rhode Island

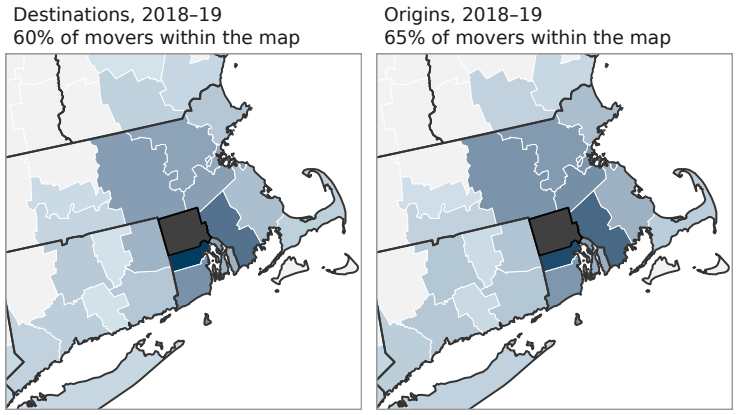
Used as a placebo: passes both pre-2020 screens

→ = 2018–19 to 2021–22 average	Providence	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	660,741	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	1,614	1,891	Origin county identified in the ACS?	yes	
Urban share of population	95%	99%	(a) Own pre-trend, joint p-value	0.12 — pass	0.39
Median household income (ACS)	$58,974	$69,176	(b) IRS slope distance from Multnomah	0.43 — pass	cutoff 0.43
Share under 24 / 65 and over	32% / 15%	27% / 13%	rank among donors (1 = closest)	10 of 20	
State top income-tax rate, 2019 / 2020 (%)	5.99 / 5.99	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.18 / -0.06	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	42	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	5.11 → 5.40	7.01 → 9.69	DiD, out-migration: pp (se)	0.03 (0.59)	1.88 (0.57)
IRS AGI in-migration rate (%)	3.99 → 4.61	6.59 → 6.29	DiD, out-of-state: pp (se)	0.22 (0.51)	1.34 (0.44)
IRS AGI net migration rate (%)	-1.11 → -0.78	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0004 (0.0009)	-0.0028 (0.0020)

Migration rates before and after 2020



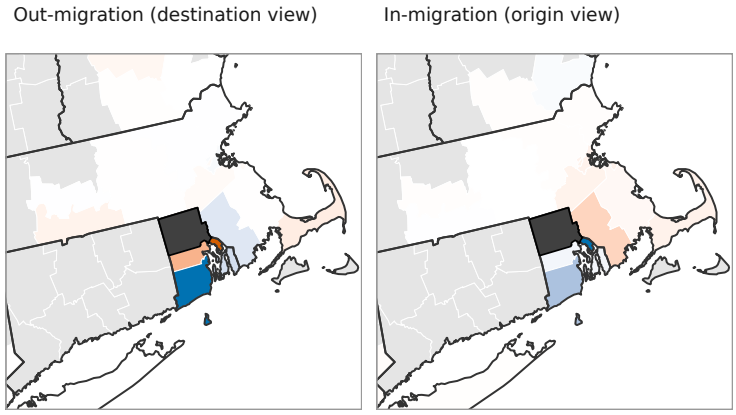
Where its movers went and came from



% of the county's domestic movers (returns)

0 5 10 15

Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)

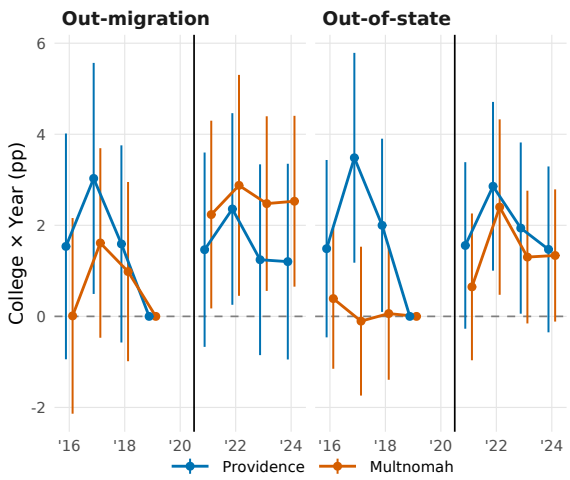


Rate change per 100,000 of partner-county AGI (scale capped)

0

% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Kent, RI	19.6	18.3	Border county	41	36	46	43
Bristol, MA	9.0	7.6	Within state	29	28	47	45
Washington, RI	4.8	4.7	Bordering state	33	29	27	27
Worcester, MA	3.6	3.4	Top 10 flows	53	49	56	54
Norfolk, MA	3.5	3.2	**In-movers, from:**				
Top origins			Border county	41	38	35	34
Kent, RI	16.4	14.7	Within state	26	24	39	39
Bristol, MA	10.5	10.3	Bordering state	42	43	28	30
Norfolk, MA	5.1	5.2	Top 10 flows	57	55	48	48
Worcester, MA	4.6	4.2					
Suffolk, MA	4.4	5.2					

Placebo event study (base 2019, 95% CI)



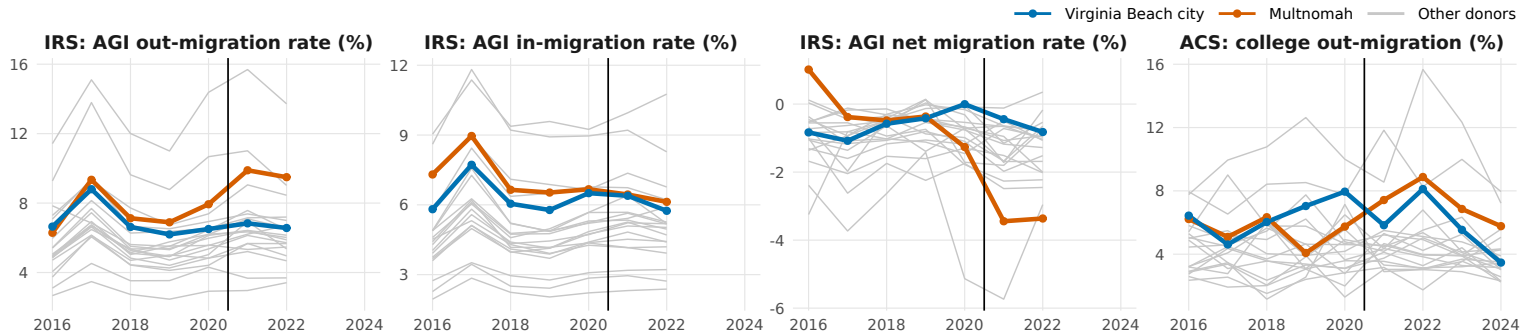
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Virginia Beach city, Virginia

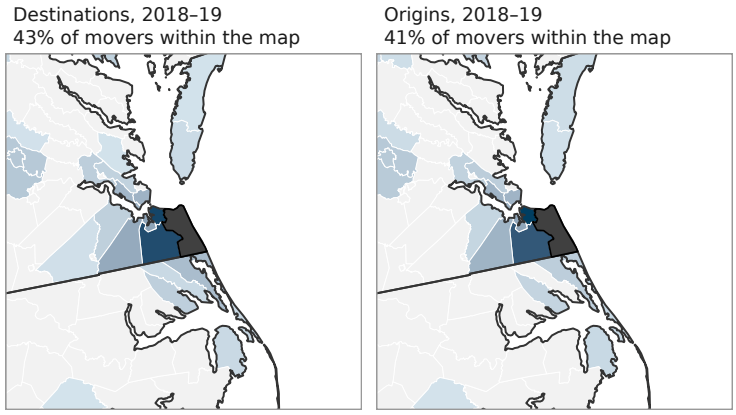
Used as a placebo: passes both pre-2020 screens

→ = 2018–19 to 2021–22 average	Virginia Beach city	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	459,470	815,428	Used as a placebo (both screens)?	YES	
Density (people per sq. mile)	1,878	1,891	Origin county identified in the ACS?	yes	
Urban share of population	98%	99%	(a) Own pre-trend, joint p-value	0.21 — pass	0.39
Median household income (ACS)	$76,610	$69,176	(b) IRS slope distance from Multnomah	0.42 — pass	cutoff 0.43
Share under 24 / 65 and over	32% / 14%	27% / 13%	rank among donors (1 = closest)	9 of 20	
State top income-tax rate, 2019 / 2020 (%)	5.75 / 5.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.35 / -0.18	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	46	88	Other donor pools	urban-COVID; demog.	
IRS AGI out-migration rate (%)	6.41 → 6.70	7.01 → 9.69	DiD, out-migration: pp (se)	-0.08 (0.76)	1.88 (0.57)
IRS AGI in-migration rate (%)	5.91 → 6.07	6.59 → 6.29	DiD, out-of-state: pp (se)	0.24 (0.58)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.50 → -0.63	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0008 (0.0013)	-0.0028 (0.0020)

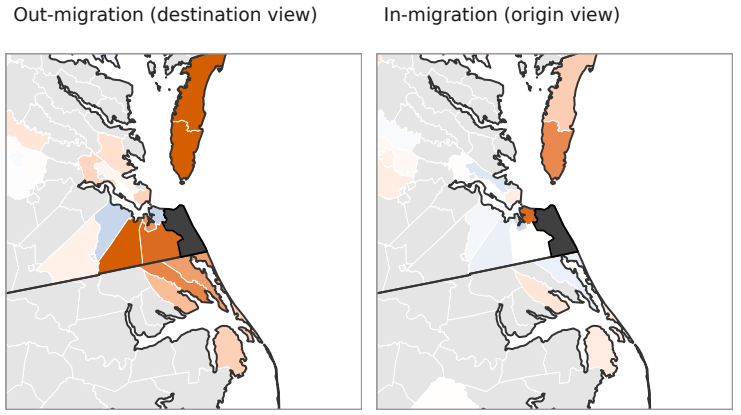
Migration rates before and after 2020



Where its movers went and came from

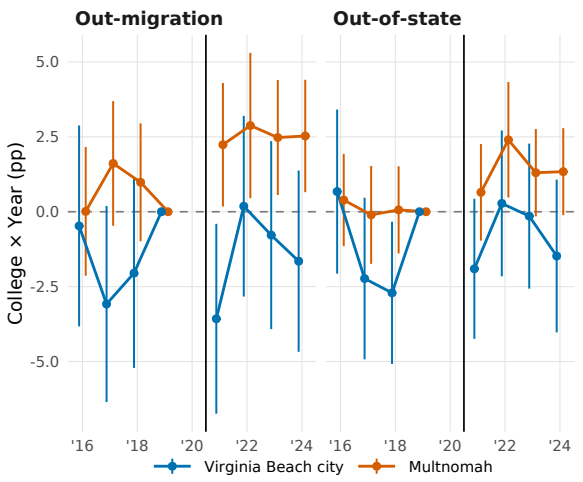


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Norfolk, VA	15.2	14.6	Border county	32	32	46	43
Chesapeake, VA	13.8	14.6	Within state	47	50	47	45
Portsmouth, VA	2.6	3.2	Bordering state	7	8	27	27
San Diego, CA	2.5	2.1	Top 10 flows	43	45	56	54
Suffolk, VA	2.2	3.0	**In-movers, from:**				
Top origins			Border county	31	31	35	34
Norfolk, VA	16.8	16.7	Within state	46	47	39	39
Chesapeake, VA	11.7	12.1	Bordering state	6	6	28	30
San Diego, CA	2.9	2.6	Top 10 flows	42	42	48	48
Portsmouth, VA	2.4	2.5					
Newport News, VA	1.6	1.5					

Placebo event study (base 2019, 95% CI)



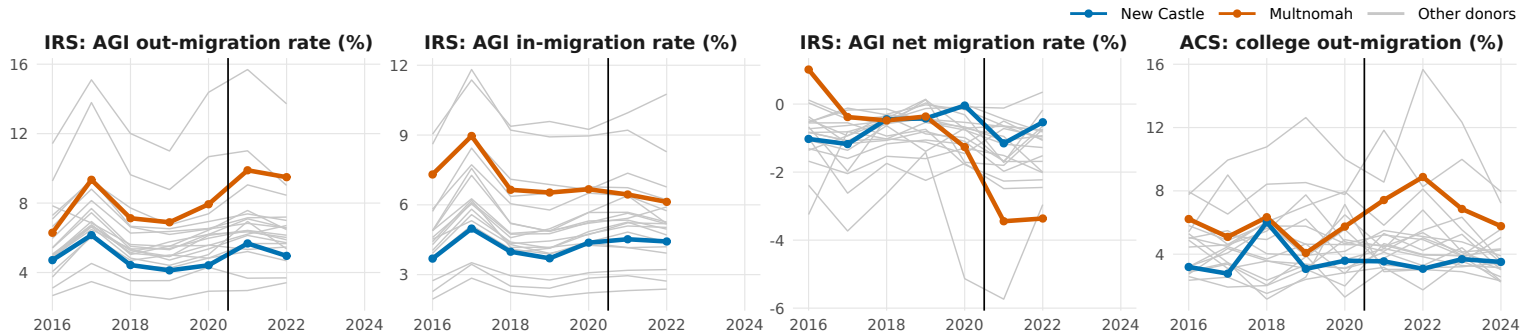
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

New Castle County, Delaware

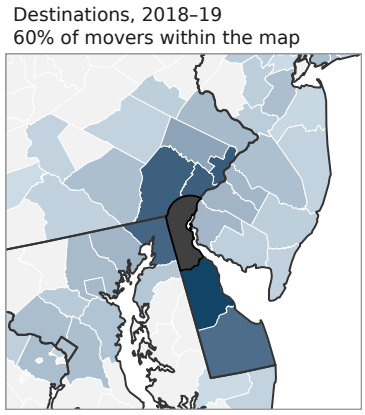
Not used as a placebo: fails the IRS-trend screen (b)

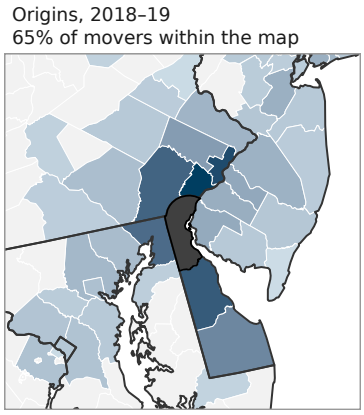
→ = 2018–19 to 2021–22 average	New Castle	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	570,719	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	1,339	1,891	Origin county identified in the ACS?	yes	
Urban share of population	94%	99%	(a) Own pre-trend, joint p-value	0.14 — pass	0.39
Median household income (ACS)	$73,892	$69,176	(b) IRS slope distance from Multnomah	0.48 — FAIL	cutoff 0.43
Share under 24 / 65 and over	31% / 15%	27% / 13%	rank among donors (1 = closest)	13 of 20	
State top income-tax rate, 2019 / 2020 (%)	6.60 / 6.60	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.35 / -0.10	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	59	88	Other donor pools	demog.	
IRS AGI out-migration rate (%)	4.28 → 5.32	7.01 → 9.69	DiD, out-migration: pp (se)	0.60 (0.60)	1.88 (0.57)
IRS AGI in-migration rate (%)	3.84 → 4.47	6.59 → 6.29	DiD, out-of-state: pp (se)	0.85 (0.56)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.43 → -0.84	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0000 (0.0014)	-0.0028 (0.0020)

Migration rates before and after 2020

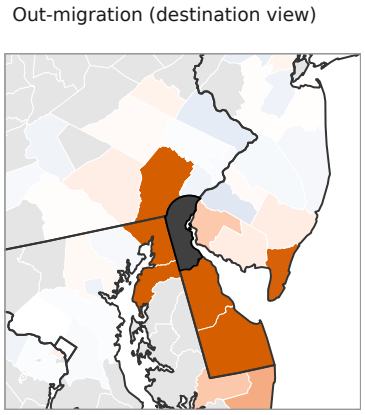


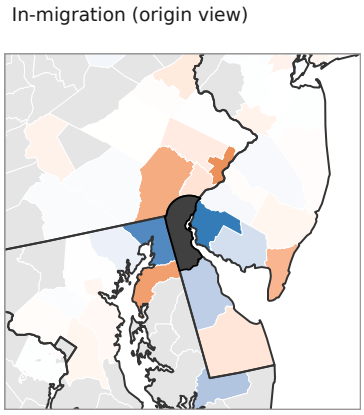
Where its movers went and came from

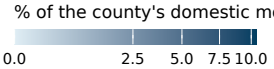


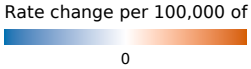


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



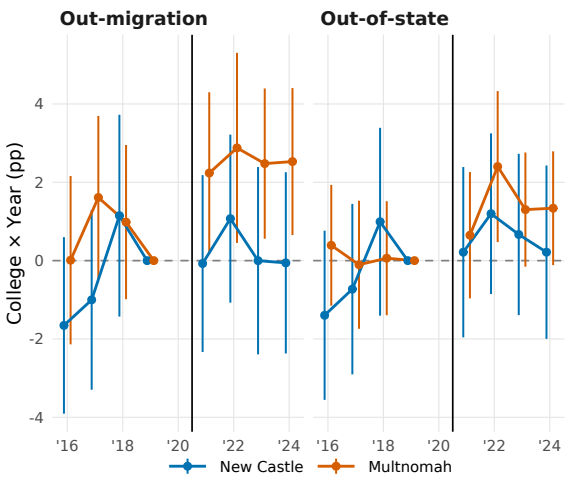






% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Kent, DE	9.5	9.8	Border county	30	30	46	43
Philadelphia, PA	6.8	6.1	Within state	15	16	47	45
Delaware, PA	6.5	5.8	Bordering state	43	40	27	27
Chester, PA	6.4	5.9	Top 10 flows	45	44	56	54
Cecil, MD	5.9	6.0	**In-movers, from:**				
Top origins			Border county	32	30	35	34
Delaware, PA	10.4	10.4	Within state	10	10	39	39
Philadelphia, PA	8.7	11.4	Bordering state	52	52	28	30
Kent, DE	7.0	6.7	Top 10 flows	47	48	48	48
Chester, PA	6.0	5.5					
Cecil, MD	5.3	4.4					

Placebo event study (base 2019, 95% CI)



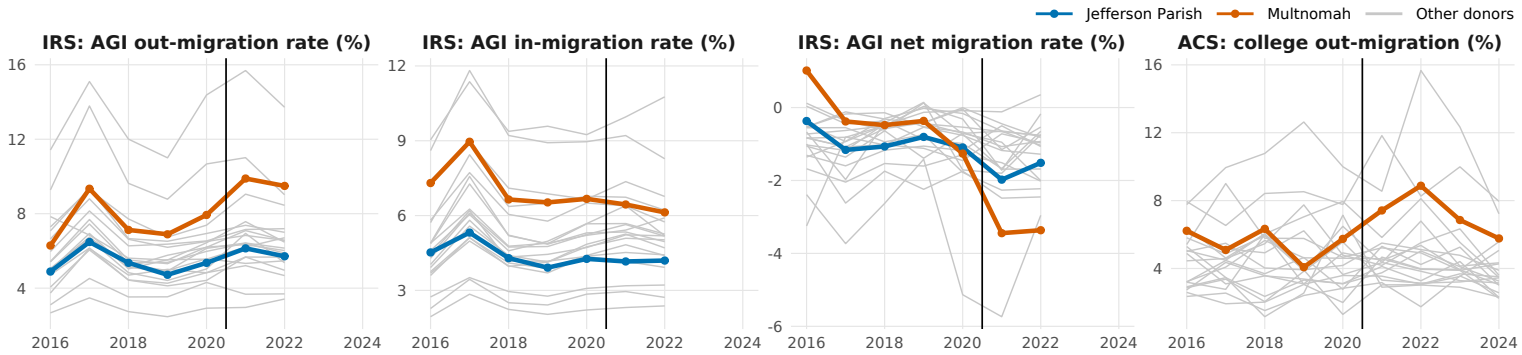
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Jefferson Parish, Louisiana

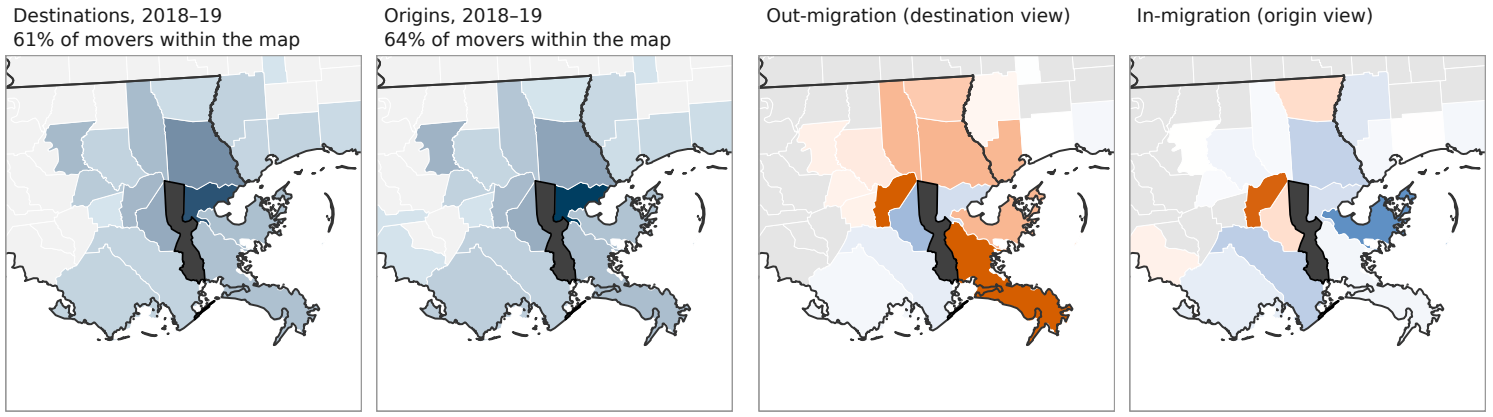
Not used as a placebo: not identified in the ACS

→ = 2018–19 to 2021–22 average	Jefferson Parish	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	440,781	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	1,465	1,891	Origin county identified in the ACS?	NO	
Urban share of population	98%	99%	(a) Own pre-trend, joint p-value	n/a	0.39
Median household income (ACS)	$54,032	$69,176	(b) IRS slope distance from Multnomah	0.22 — pass	cutoff 0.43
Share under 24 / 65 and over	30% / 17%	27% / 13%	rank among donors (1 = closest)	1 of 20	
State top income-tax rate, 2019 / 2020 (%)	6.00 / 6.00	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.16 / -0.29	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	53	88	Other donor pools	none	
IRS AGI out-migration rate (%)	5.04 → 5.93	7.01 → 9.69	DiD, out-migration: pp (se)	n/a	1.88 (0.57)
IRS AGI in-migration rate (%)	4.10 → 4.18	6.59 → 6.29	DiD, out-of-state: pp (se)	n/a	1.34 (0.44)
IRS AGI net migration rate (%)	-0.94 → -1.75	-0.42 → -3.40	DiD, in-migration: pp (se)	n/a	-0.0028 (0.0020)

Migration rates before and after 2020



Where its movers went and came from



Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Orleans, LA	27.6	23.3	Border county	50	47	46	43
St. Tammany, LA	9.8	11.0	Within state	62	60	47	45
St. Charles, LA	4.7	4.2	Bordering state	10	12	27	27
Harris, TX	3.3	4.1	Top 10 flows	59	57	56	54
St. John the Baptist, LA	3.0	3.0	**In-movers, from:**				
Top origins			Border county	54	56	35	34
Orleans, LA	36.8	38.0	Within state	67	68	39	39
St. Tammany, LA	5.6	5.5	Bordering state	7	6	28	30
St. Charles, LA	4.3	4.6	Top 10 flows	62	64	48	48
East Baton Rouge, LA	3.5	3.1					
St. John the Baptist, LA	2.7	3.1					

Not identified as an origin county in the ACS public-use file, so no individual-level placebo DiD can be run for this county.

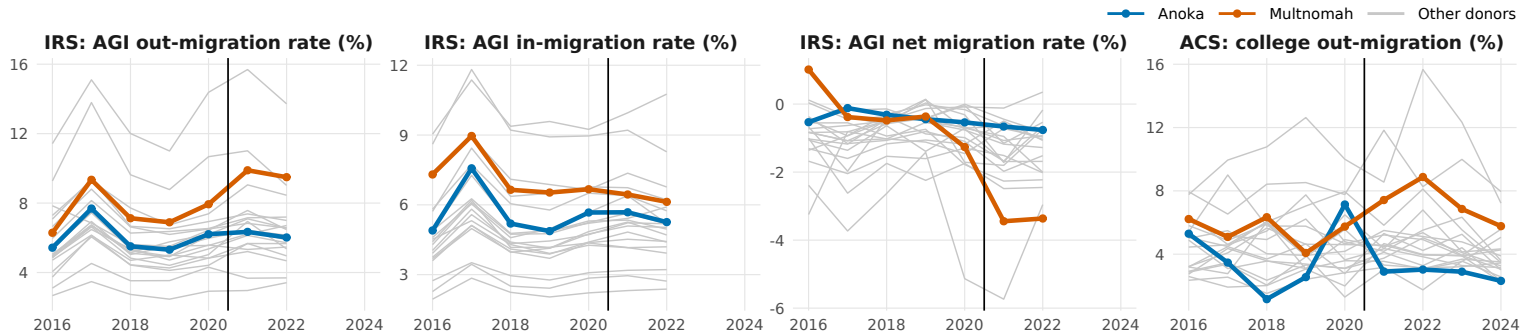
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Anoka County, Minnesota

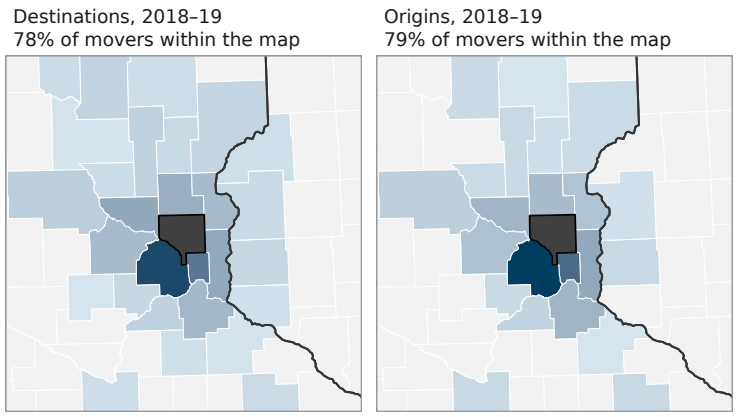
Not used as a placebo: fails the own pre-trend screen (a)

→ = 2018–19 to 2021–22 average	Anoka	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	363,887	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	862	1,891	Origin county identified in the ACS?	yes	
Urban share of population	84%	99%	(a) Own pre-trend, joint p-value	0.07 — FAIL	0.39
Median household income (ACS)	$82,175	$69,176	(b) IRS slope distance from Multnomah	0.30 — pass	cutoff 0.43
Share under 24 / 65 and over	32% / 14%	27% / 13%	rank among donors (1 = closest)	3 of 20	
State top income-tax rate, 2019 / 2020 (%)	9.85 / 9.85	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.25 / -0.25	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	51	88	Other donor pools	stringency	
IRS AGI out-migration rate (%)	5.41 → 6.18	7.01 → 9.69	DiD, out-migration: pp (se)	0.34 (0.83)	1.88 (0.57)
IRS AGI in-migration rate (%)	5.03 → 5.47	6.59 → 6.29	DiD, out-of-state: pp (se)	0.14 (0.31)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.38 → -0.71	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0001 (0.0007)	-0.0028 (0.0020)

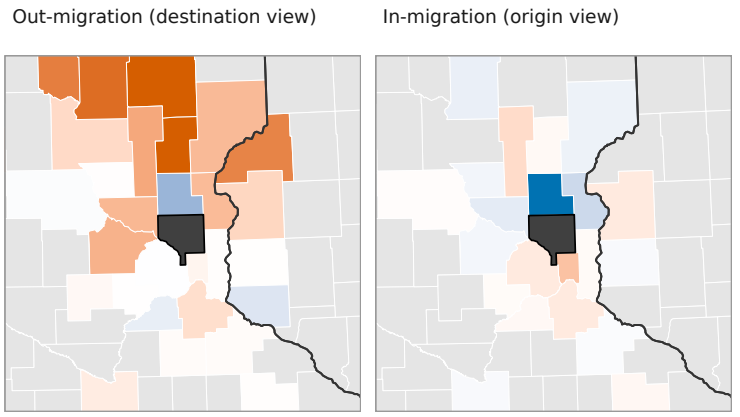
Migration rates before and after 2020



Where its movers went and came from

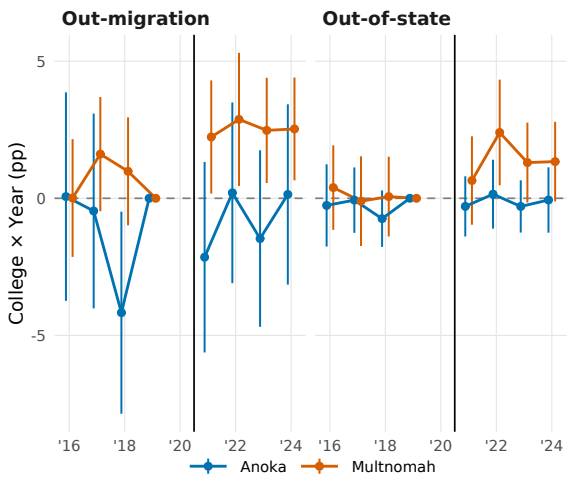


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Hennepin, MN	31.4	28.6	Border county	63	59	46	43
Ramsey, MN	15.2	14.0	Within state	80	78	47	45
Sherburne, MN	5.2	5.2	Bordering state	2	2	27	27
Washington, MN	5.0	4.8	Top 10 flows	71	67	56	54
Isanti, MN	4.2	3.7	**In-movers, from:**				
Top origins			Border county	68	69	35	34
Hennepin, MN	36.3	37.5	Within state	83	84	39	39
Ramsey, MN	19.1	20.2	Bordering state	2	1	28	30
Washington, MN	5.0	4.9	Top 10 flows	75	77	48	48
Dakota, MN	3.4	3.5					
Sherburne, MN	3.3	3.2					

Placebo event study (base 2019, 95% CI)



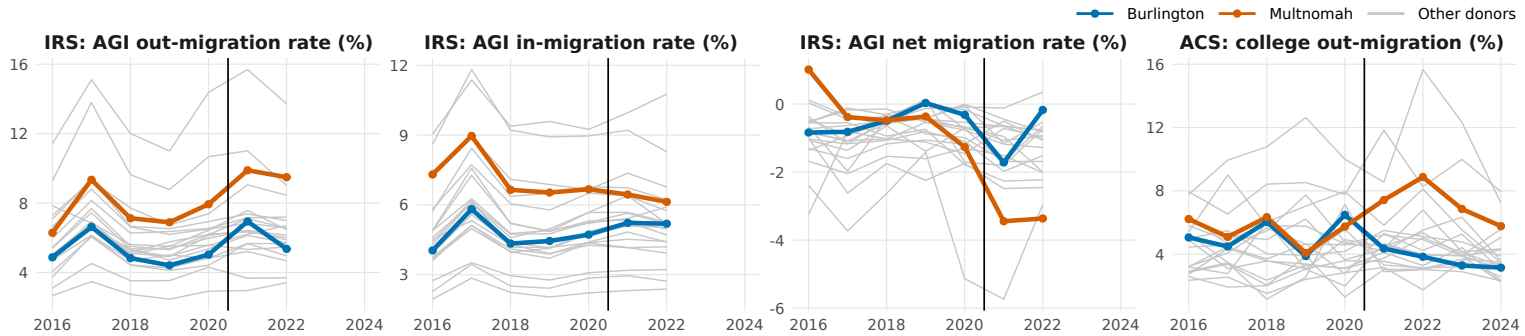
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Burlington County, New Jersey

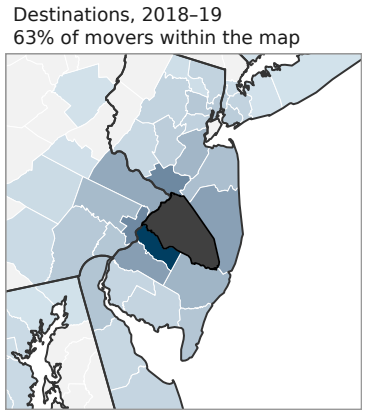
Not used as a placebo: fails the IRS-trend screen (b)

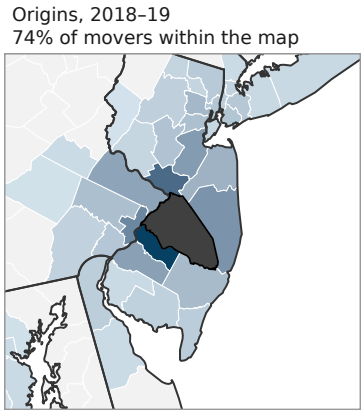
→ = 2018–19 to 2021–22 average	Burlington	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	461,860	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	578	1,891	Origin county identified in the ACS?	yes	
Urban share of population	90%	99%	(a) Own pre-trend, joint p-value	0.39 — pass	0.39
Median household income (ACS)	$87,416	$69,176	(b) IRS slope distance from Multnomah	0.52 — FAIL	cutoff 0.43
Share under 24 / 65 and over	30% / 17%	27% / 13%	rank among donors (1 = closest)	16 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.32 / -0.03	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	4.62 → 6.15	7.01 → 9.69	DiD, out-migration: pp (se)	-0.35 (0.63)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.39 → 5.21	6.59 → 6.29	DiD, out-of-state: pp (se)	-0.80 (0.48)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.23 → -0.95	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0013 (0.0009)	-0.0028 (0.0020)

Migration rates before and after 2020

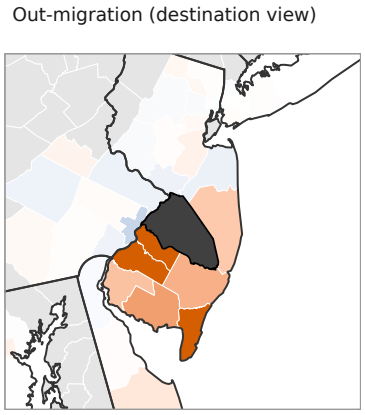


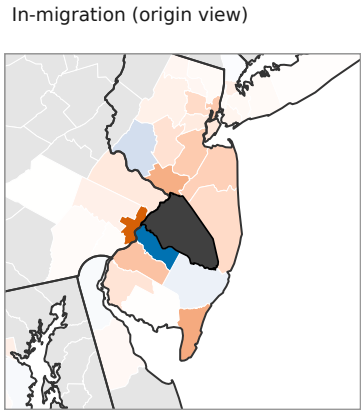
Where its movers went and came from





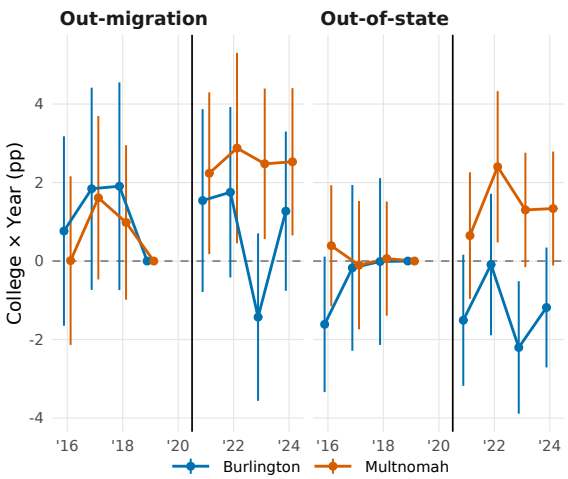
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Camden, NJ	21.0	19.3	Border county	43	40	46	43
Mercer, NJ	6.3	5.6	Within state	45	45	47	45
Philadelphia, PA	6.2	5.2	Bordering state	17	16	27	27
Gloucester, NJ	3.8	4.6	Top 10 flows	50	48	56	54
Ocean, NJ	3.6	3.5	**In-movers, from:**				
Top origins			Border county	50	49	35	34
Camden, NJ	20.4	17.9	Within state	56	56	39	39
Mercer, NJ	11.0	11.0	Bordering state	17	20	28	30
Philadelphia, PA	6.5	8.0	Top 10 flows	59	59	48	48
Ocean, NJ	4.9	5.1					
Middlesex, NJ	4.0	4.6					

Placebo event study (base 2019, 95% CI)



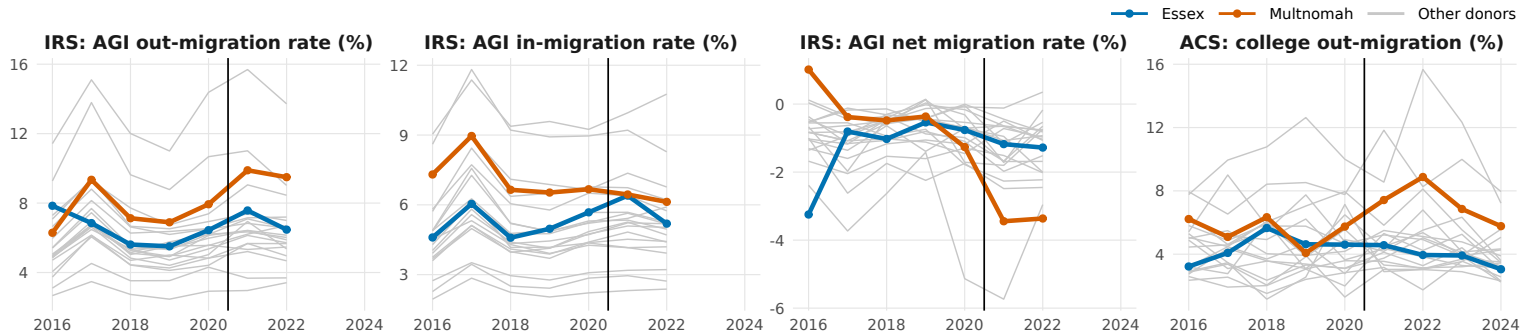
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Essex County, New Jersey

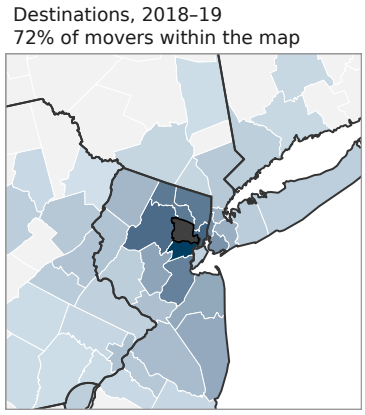
Not used as a placebo: fails the IRS-trend screen (b)

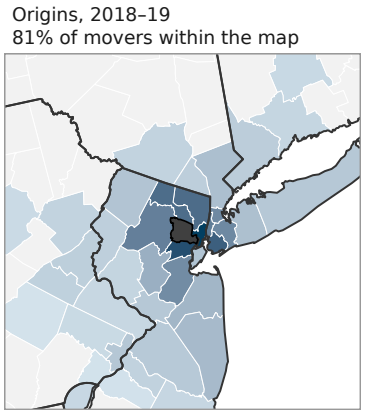
→ = 2018–19 to 2021–22 average	Essex	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	863,728	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	6,850	1,891	Origin county identified in the ACS?	yes	
Urban share of population	100%	99%	(a) Own pre-trend, joint p-value	0.55 — pass	0.39
Median household income (ACS)	$61,510	$69,176	(b) IRS slope distance from Multnomah	0.89 — FAIL	cutoff 0.43
Share under 24 / 65 and over	33% / 13%	27% / 13%	rank among donors (1 = closest)	20 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.82 / -0.03	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	5.56 → 7.02	7.01 → 9.69	DiD, out-migration: pp (se)	-0.64 (0.52)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.78 → 5.79	6.59 → 6.29	DiD, out-of-state: pp (se)	-0.23 (0.37)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.78 → -1.23	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0020 (0.0017)	-0.0028 (0.0020)

Migration rates before and after 2020

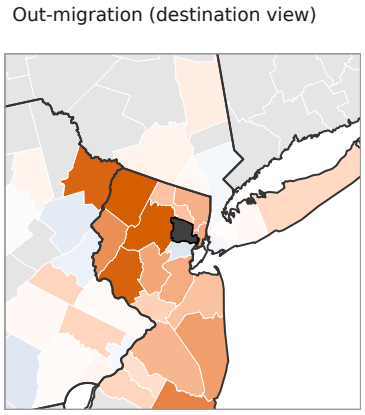


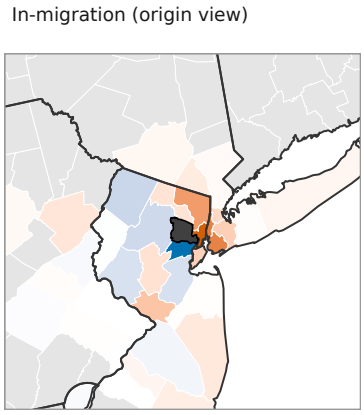
Where its movers went and came from

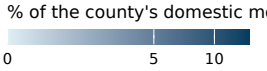


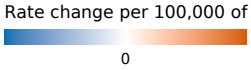


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



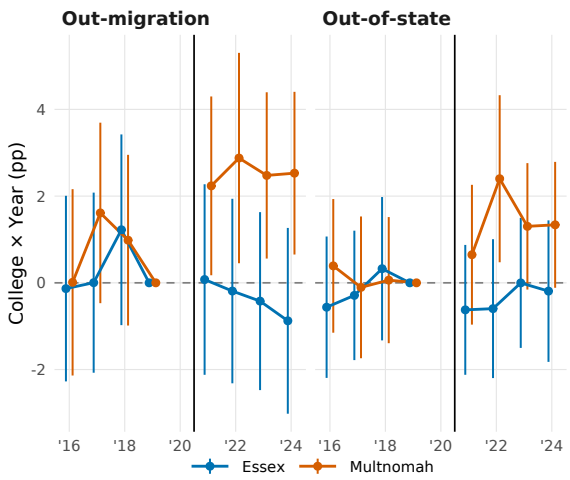






% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Union, NJ	13.6	11.9	Border county	39	37	46	43
Hudson, NJ	7.8	7.7	Within state	55	55	47	45
Morris, NJ	7.1	7.5	Bordering state	17	15	27	27
Passaic, NJ	5.4	5.0	Top 10 flows	55	52	56	54
Bergen, NJ	5.4	5.1	**In-movers, from:**				
Top origins			Border county	43	41	35	34
Hudson, NJ	13.5	13.4	Within state	53	51	39	39
Union, NJ	10.8	9.4	Bordering state	28	30	28	30
Kings, NY	8.5	9.3	Top 10 flows	68	68	48	48
Bergen, NJ	6.9	7.0					
Passaic, NJ	6.8	7.2					

Placebo event study (base 2019, 95% CI)



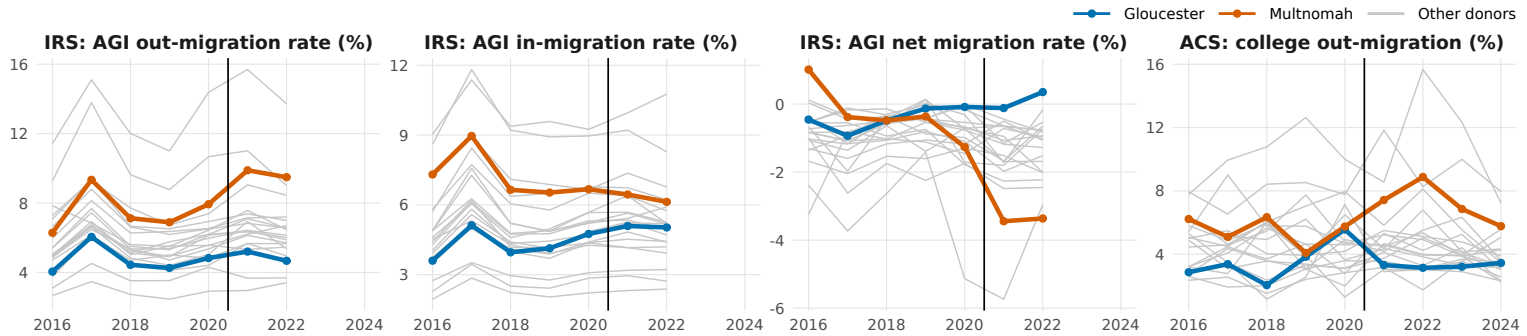
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Gloucester County, New Jersey

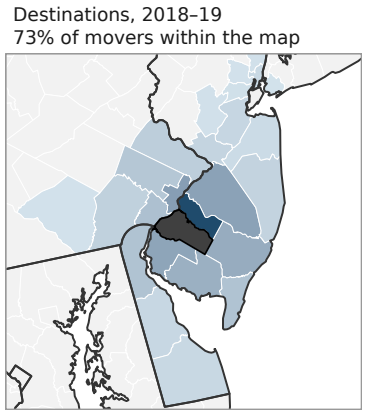
Not used as a placebo: fails the own pre-trend screen (a); fails the IRS-trend screen (b)

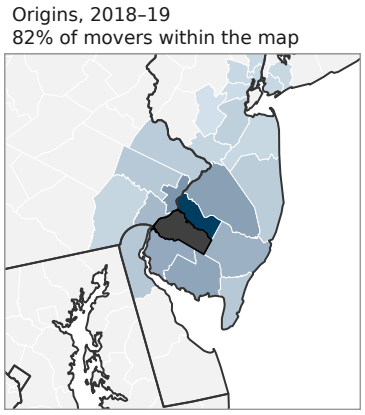
→ = 2018–19 to 2021–22 average	Gloucester	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	302,294	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	939	1,891	Origin county identified in the ACS?	yes	
Urban share of population	89%	99%	(a) Own pre-trend, joint p-value	0.02 — FAIL	0.39
Median household income (ACS)	$87,283	$69,176	(b) IRS slope distance from Multnomah	0.51 — FAIL	cutoff 0.43
Share under 24 / 65 and over	31% / 15%	27% / 13%	rank among donors (1 = closest)	15 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.10 / 0.04	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency	
IRS AGI out-migration rate (%)	4.35 → 4.94	7.01 → 9.69	DiD, out-migration: pp (se)	1.08 (0.76)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.05 → 5.06	6.59 → 6.29	DiD, out-of-state: pp (se)	0.91 (0.52)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.31 → 0.12	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0004 (0.0006)	-0.0028 (0.0020)

Migration rates before and after 2020

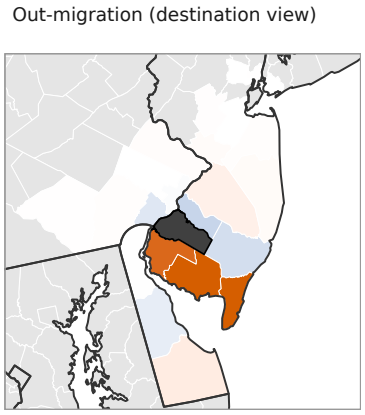


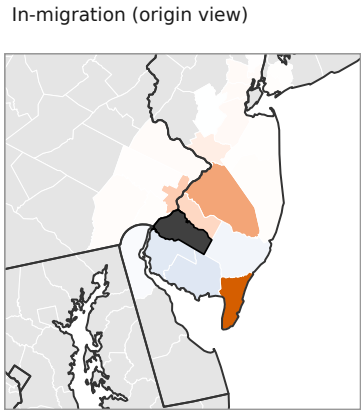
Where its movers went and came from





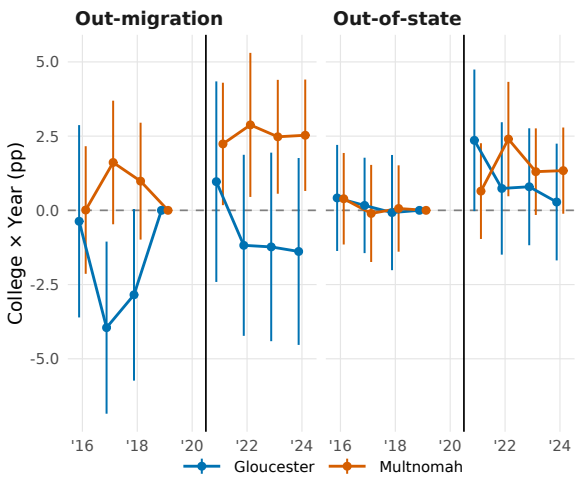
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Camden, NJ	31.6	28.4	Border county	57	54	46	43
Philadelphia, PA	6.8	6.4	Within state	58	57	47	45
Burlington, NJ	6.2	6.0	Bordering state	16	15	27	27
Salem, NJ	5.7	5.9	Top 10 flows	66	64	56	54
Atlantic, NJ	4.5	4.5	**In-movers, from:**				
Top origins			Border county	65	61	35	34
Camden, NJ	37.6	33.3	Within state	66	62	39	39
Philadelphia, PA	8.3	11.6	Bordering state	17	21	28	30
Burlington, NJ	6.5	7.4	Top 10 flows	74	72	48	48
Cumberland, NJ	5.6	4.6					
Salem, NJ	5.0	4.0					

Placebo event study (base 2019, 95% CI)



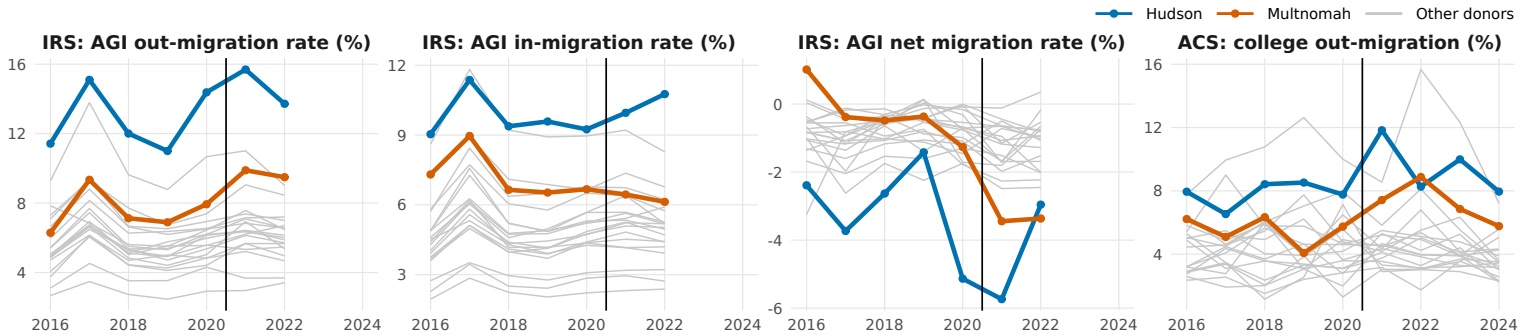
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Hudson County, New Jersey

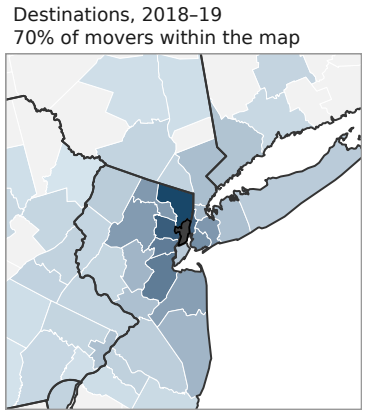
Not used as a placebo: fails the IRS-trend screen (b)

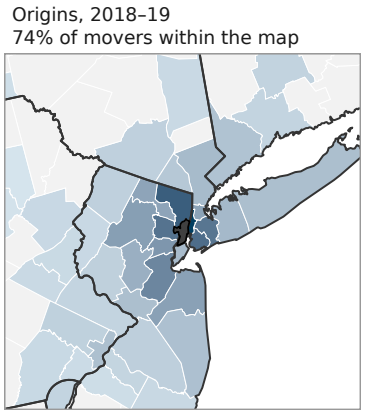
→ = 2018–19 to 2021–22 average	Hudson	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	724,854	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	15,692	1,891	Origin county identified in the ACS?	yes	
Urban share of population	100%	99%	(a) Own pre-trend, joint p-value	0.44 — pass	0.39
Median household income (ACS)	$71,189	$69,176	(b) IRS slope distance from Multnomah	0.58 — FAIL	cutoff 0.43
Share under 24 / 65 and over	29% / 12%	27% / 13%	rank among donors (1 = closest)	17 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.44 / -0.04	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	11.50 → 14.70	7.01 → 9.69	DiD, out-migration: pp (se)	0.35 (0.68)	1.88 (0.57)
IRS AGI in-migration rate (%)	9.48 → 10.36	6.59 → 6.29	DiD, out-of-state: pp (se)	0.01 (0.54)	1.34 (0.44)
IRS AGI net migration rate (%)	-2.02 → -4.35	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0000 (0.0015)	-0.0028 (0.0020)

Migration rates before and after 2020

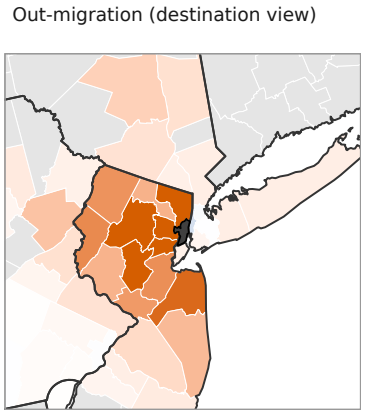


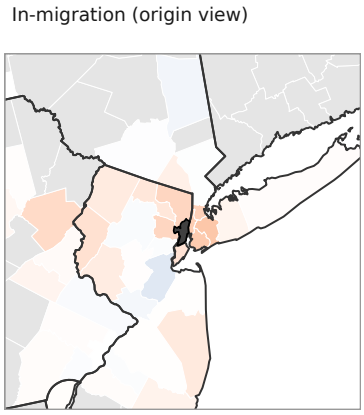
Where its movers went and came from





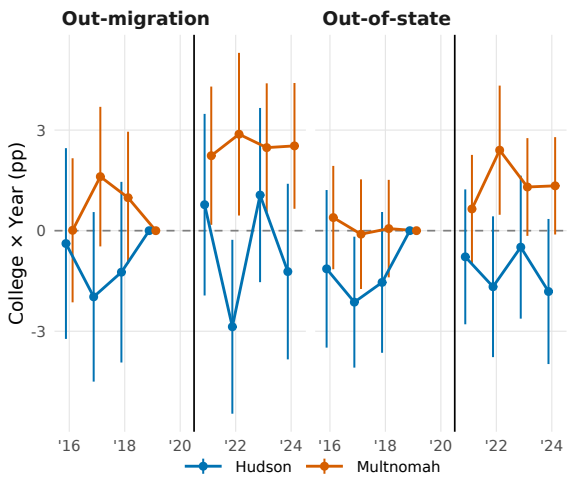
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Bergen, NJ	12.7	11.9	Border county	39	34	46	43
Essex, NJ	9.3	8.2	Within state	47	48	47	45
New York, NY	7.7	5.3	Bordering state	23	20	27	27
Middlesex, NJ	5.5	5.8	Top 10 flows	56	52	56	54
Union, NJ	5.5	5.5	**In-movers, from:**				
Top origins			Border county	41	42	35	34
New York, NY	14.3	14.3	Within state	35	33	39	39
Bergen, NJ	9.1	8.4	Bordering state	39	42	28	30
Kings, NY	7.1	8.5	Top 10 flows	58	59	48	48
Essex, NJ	6.4	6.3					
Queens, NY	6.2	6.9					

Placebo event study (base 2019, 95% CI)



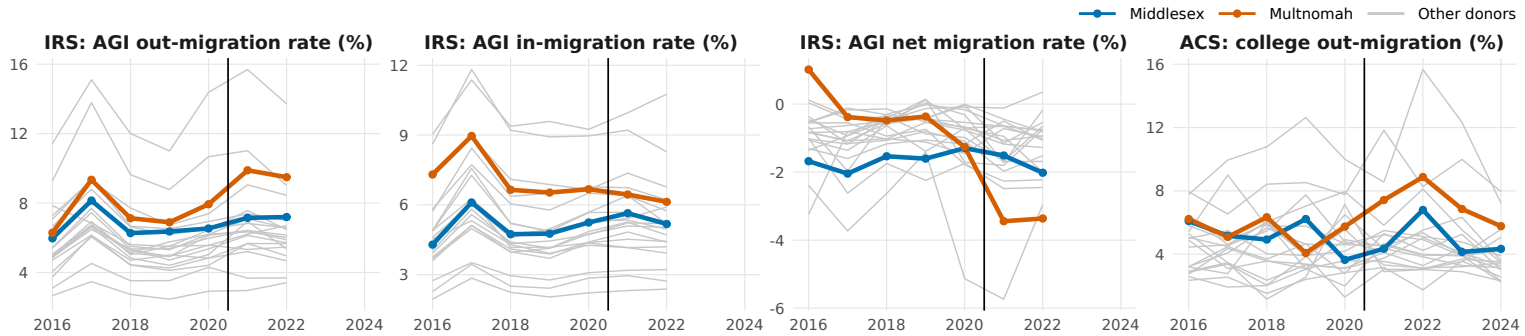
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Middlesex County, New Jersey

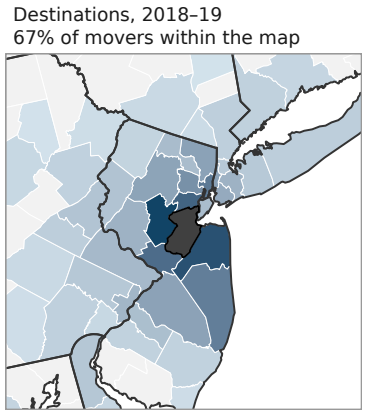
Not used as a placebo: fails the IRS-trend screen (b)

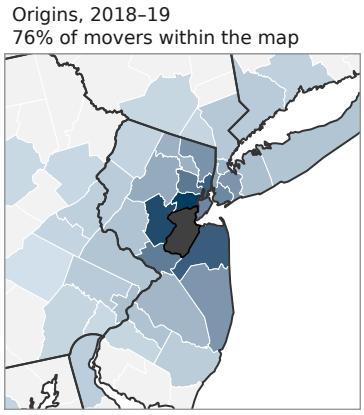
→ = 2018–19 to 2021–22 average	Middlesex	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	863,162	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	2,791	1,891	Origin county identified in the ACS?	yes	
Urban share of population	99%	99%	(a) Own pre-trend, joint p-value	0.45 — pass	0.39
Median household income (ACS)	$89,533	$69,176	(b) IRS slope distance from Multnomah	0.47 — FAIL	cutoff 0.43
Share under 24 / 65 and over	31% / 15%	27% / 13%	rank among donors (1 = closest)	12 of 20	
State top income-tax rate, 2019 / 2020 (%)	10.75 / 10.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.07 / 0.00	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	80	88	Other donor pools	stringency; demog.	
IRS AGI out-migration rate (%)	6.32 → 7.17	7.01 → 9.69	DiD, out-migration: pp (se)	0.24 (0.50)	1.88 (0.57)
IRS AGI in-migration rate (%)	4.75 → 5.41	6.59 → 6.29	DiD, out-of-state: pp (se)	0.02 (0.33)	1.34 (0.44)
IRS AGI net migration rate (%)	-1.57 → -1.77	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0013 (0.0019)	-0.0028 (0.0020)

Migration rates before and after 2020

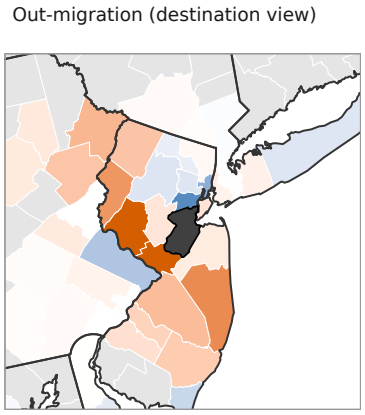


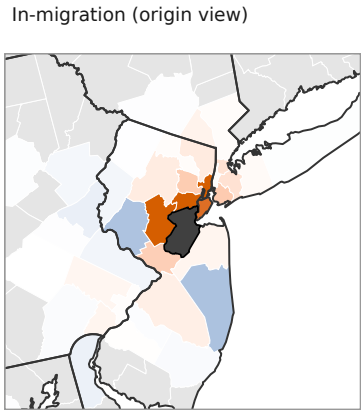
Where its movers went and came from





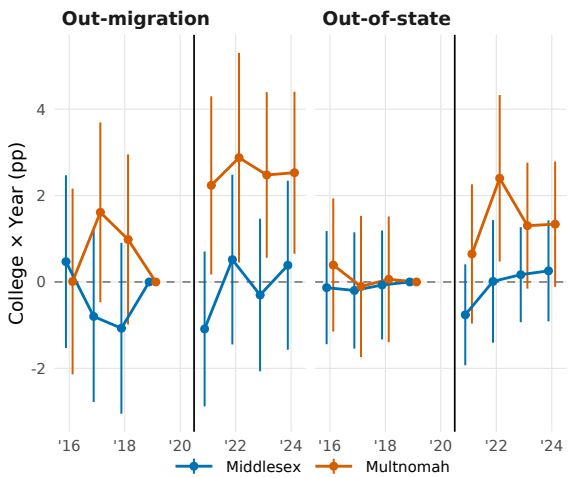
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Somerset, NJ	10.9	10.9	Border county	33	32	46	43
Monmouth, NJ	9.0	8.1	Within state	54	54	47	45
Union, NJ	6.8	6.0	Bordering state	13	13	27	27
Mercer, NJ	5.9	6.0	Top 10 flows	50	49	56	54
Hudson, NJ	4.7	4.3	**In-movers, from:**				
Top origins			Border county	37	36	35	34
Union, NJ	11.8	12.2	Within state	57	58	39	39
Somerset, NJ	9.8	8.5	Bordering state	19	22	28	30
Monmouth, NJ	7.6	7.4	Top 10 flows	58	61	48	48
Hudson, NJ	7.3	8.4					
Essex, NJ	4.8	5.6					

Placebo event study (base 2019, 95% CI)



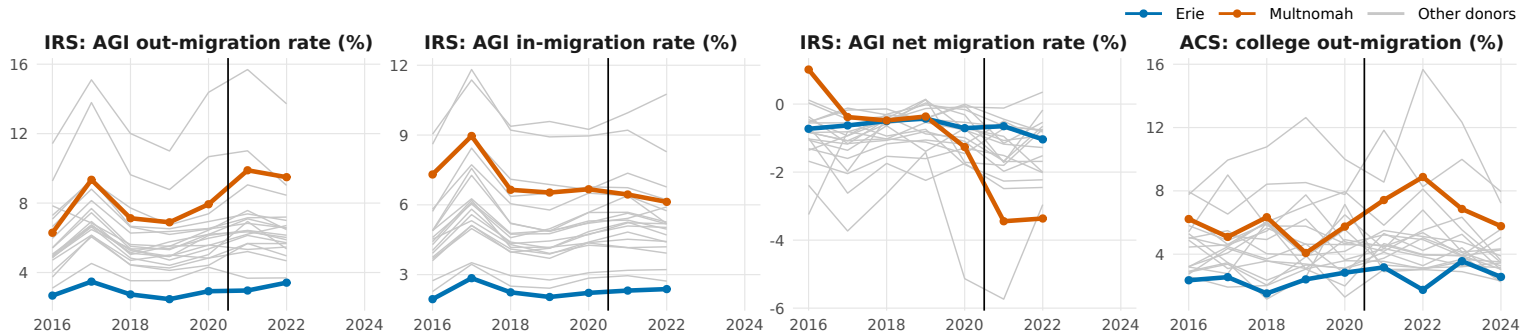
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Erie County, New York

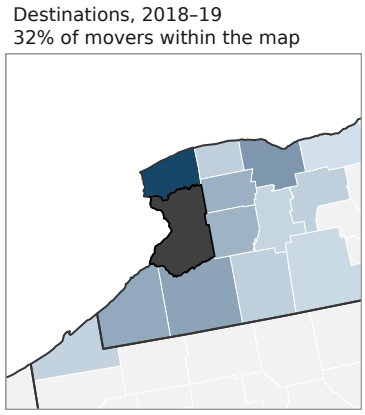
Not used as a placebo: fails the IRS-trend screen (b)

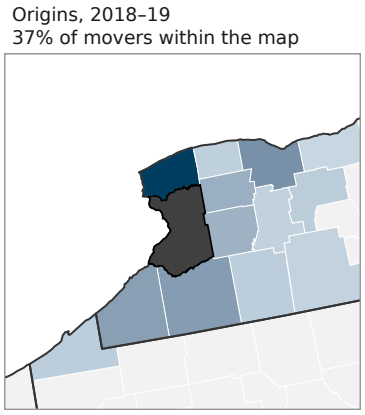
→ = 2018–19 to 2021–22 average	Erie	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	954,236	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	915	1,891	Origin county identified in the ACS?	yes	
Urban share of population	90%	99%	(a) Own pre-trend, joint p-value	0.43 — pass	0.39
Median household income (ACS)	$58,121	$69,176	(b) IRS slope distance from Multnomah	0.44 — FAIL	cutoff 0.43
Share under 24 / 65 and over	30% / 18%	27% / 13%	rank among donors (1 = closest)	11 of 20	
State top income-tax rate, 2019 / 2020 (%)	8.82 / 8.82	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.14 / -0.03	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	285	88	Other donor pools	demog.	
IRS AGI out-migration rate (%)	2.60 → 3.19	7.01 → 9.69	DiD, out-migration: pp (se)	-0.52 (0.38)	1.88 (0.57)
IRS AGI in-migration rate (%)	2.14 → 2.35	6.59 → 6.29	DiD, out-of-state: pp (se)	-0.39 (0.33)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.46 → -0.84	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0001 (0.0011)	-0.0028 (0.0020)

Migration rates before and after 2020

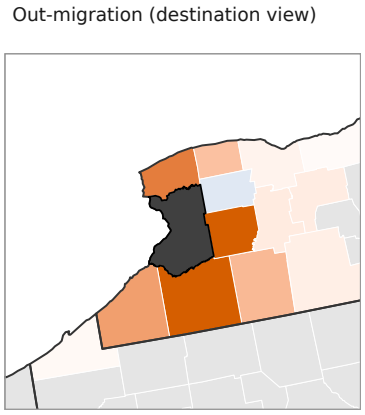


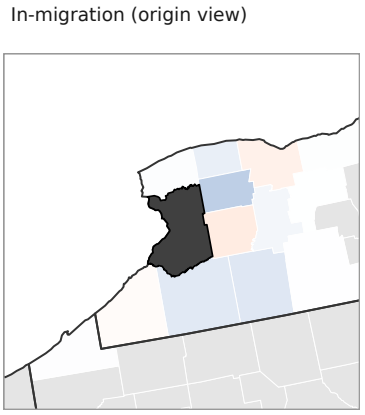
Where its movers went and came from





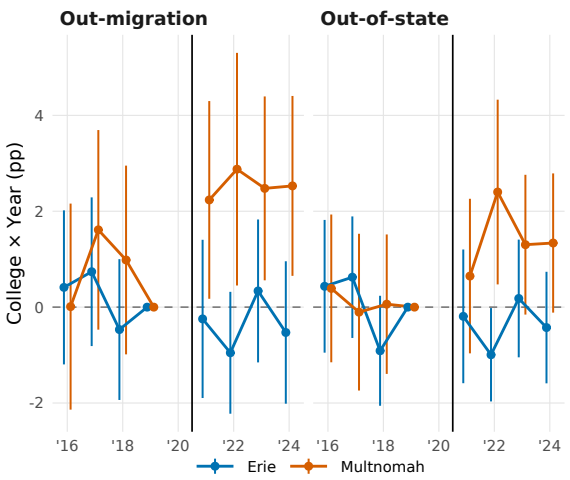
Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)





% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Niagara, NY	16.6	16.2	Border county	26	25	46	43
Monroe, NY	4.0	3.9	Within state	43	42	47	45
Cattaraugus, NY	2.9	2.9	Bordering state	4	4	27	27
Chautauqua, NY	2.5	2.5	Top 10 flows	35	34	56	54
Genesee, NY	2.0	1.8	**In-movers, from:**				
Top origins			Border county	29	25	35	34
Niagara, NY	18.6	16.2	Within state	52	54	39	39
Monroe, NY	4.6	4.6	Bordering state	4	4	28	30
Cattaraugus, NY	3.6	2.9	Top 10 flows	42	41	48	48
Chautauqua, NY	3.3	2.9					
Queens, NY	2.7	4.4					

Placebo event study (base 2019, 95% CI)



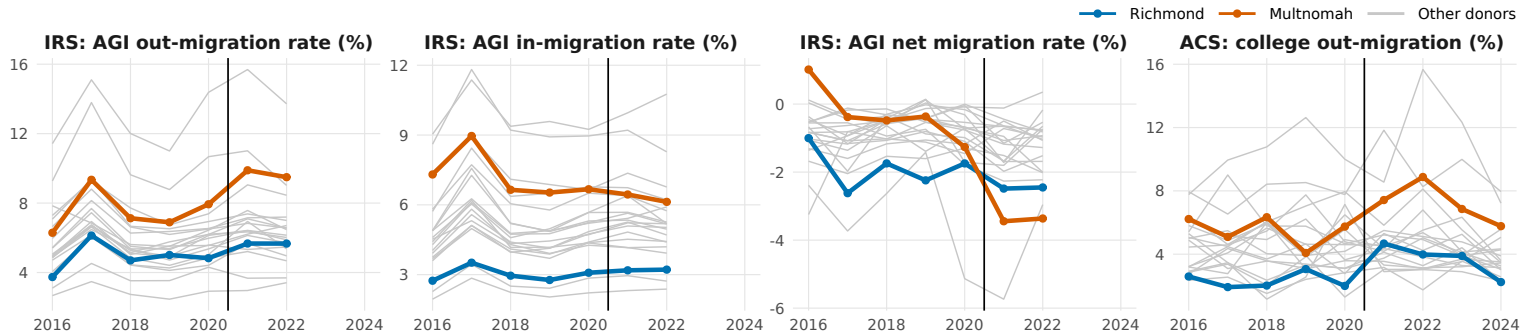
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Richmond County, New York

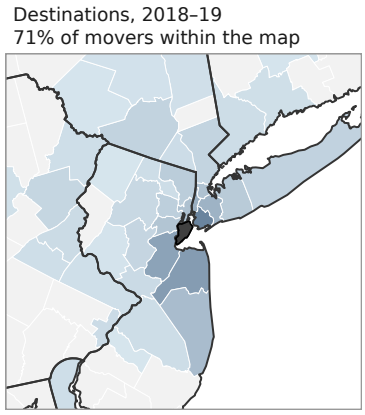
Not used as a placebo: fails the IRS-trend screen (b)

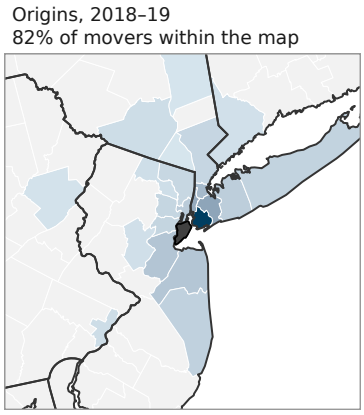
→ = 2018–19 to 2021–22 average	Richmond	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	495,747	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	8,618	1,891	Origin county identified in the ACS?	yes	
Urban share of population	100%	99%	(a) Own pre-trend, joint p-value	0.42 — pass	0.39
Median household income (ACS)	$82,783	$69,176	(b) IRS slope distance from Multnomah	0.50 — FAIL	cutoff 0.43
Share under 24 / 65 and over	30% / 16%	27% / 13%	rank among donors (1 = closest)	14 of 20	
State top income-tax rate, 2019 / 2020 (%)	8.82 / 8.82	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	0.24 / -0.05	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	YES	
Days under stay-at-home order	285	88	Other donor pools	demog.	
IRS AGI out-migration rate (%)	4.86 → 5.67	7.01 → 9.69	DiD, out-migration: pp (se)	1.02 (0.55)	1.88 (0.57)
IRS AGI in-migration rate (%)	2.86 → 3.20	6.59 → 6.29	DiD, out-of-state: pp (se)	0.70 (0.49)	1.34 (0.44)
IRS AGI net migration rate (%)	-1.99 → -2.47	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0004 (0.0006)	-0.0028 (0.0020)

Migration rates before and after 2020

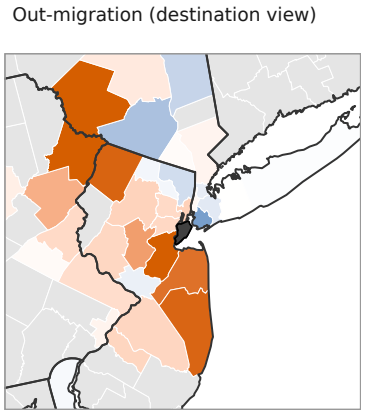


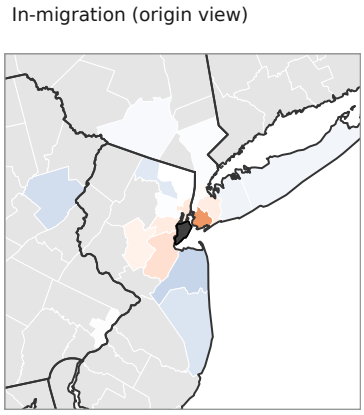
Where its movers went and came from

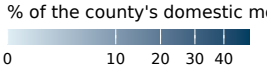


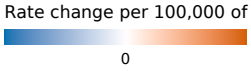


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



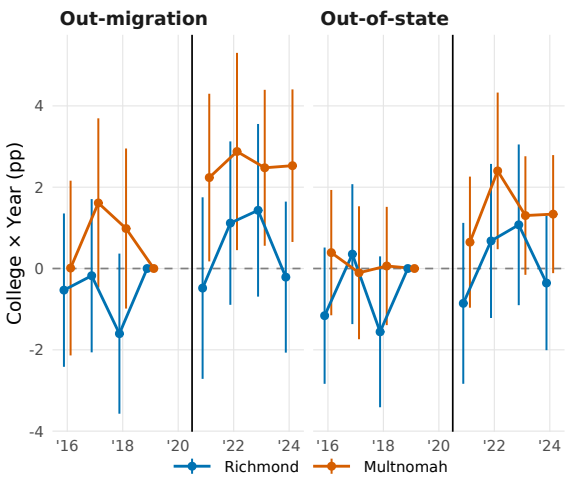






% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Kings, NY	16.2	13.1	Border county	43	40	46	43
Monmouth, NJ	9.4	8.9	Within state	37	30	47	45
Middlesex, NJ	7.9	8.4	Bordering state	35	37	27	27
New York, NY	5.5	3.9	Top 10 flows	57	51	56	54
Queens, NY	4.8	4.0	**In-movers, from:**				
Top origins			Border county	64	68	35	34
Kings, NY	49.7	53.6	Within state	73	75	39	39
Queens, NY	7.5	7.3	Bordering state	11	10	28	30
New York, NY	6.2	5.9	Top 10 flows	77	79	48	48
Bronx, NY	3.7	3.1					
Middlesex, NJ	2.2	2.1					

Placebo event study (base 2019, 95% CI)



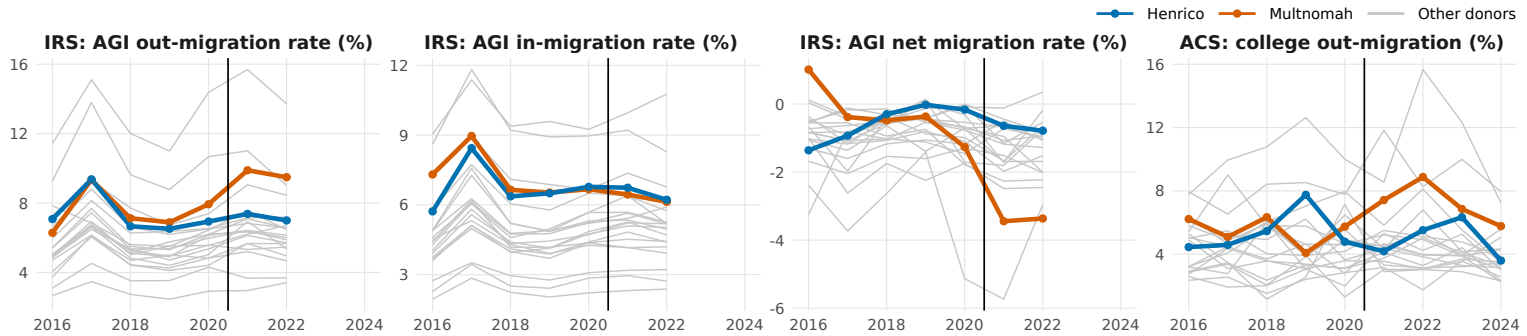
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Henrico County, Virginia

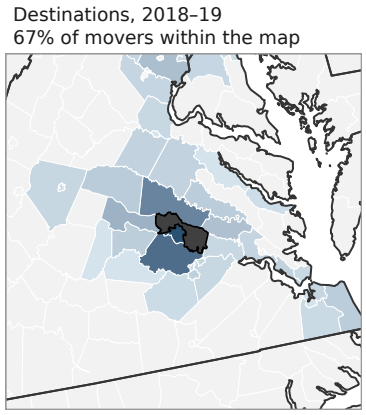
Not used as a placebo: fails the IRS-trend screen (b)

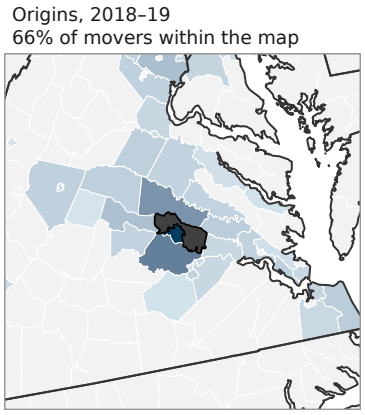
→ = 2018–19 to 2021–22 average	Henrico	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	334,389	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	1,431	1,891	Origin county identified in the ACS?	yes	
Urban share of population	96%	99%	(a) Own pre-trend, joint p-value	0.97 — pass	0.39
Median household income (ACS)	$70,307	$69,176	(b) IRS slope distance from Multnomah	0.63 — FAIL	cutoff 0.43
Share under 24 / 65 and over	31% / 15%	27% / 13%	rank among donors (1 = closest)	19 of 20	
State top income-tax rate, 2019 / 2020 (%)	5.75 / 5.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.44 / 0.03	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	46	88	Other donor pools	urban-COVID; demog.	
IRS AGI out-migration rate (%)	6.59 → 7.19	7.01 → 9.69	DiD, out-migration: pp (se)	-0.52 (0.84)	1.88 (0.57)
IRS AGI in-migration rate (%)	6.43 → 6.48	6.59 → 6.29	DiD, out-of-state: pp (se)	-0.45 (0.48)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.16 → -0.71	-0.42 → -3.40	DiD, in-migration: pp (se)	0.0011 (0.0010)	-0.0028 (0.0020)

Migration rates before and after 2020

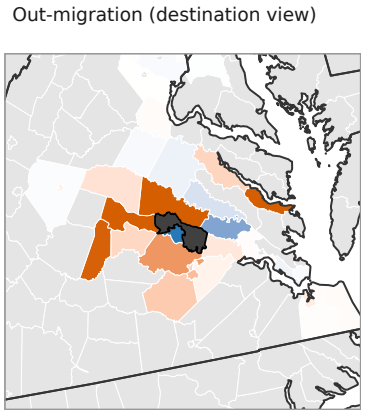


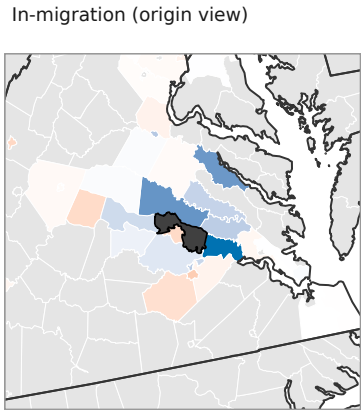
Where its movers went and came from

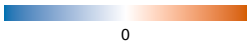




Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)

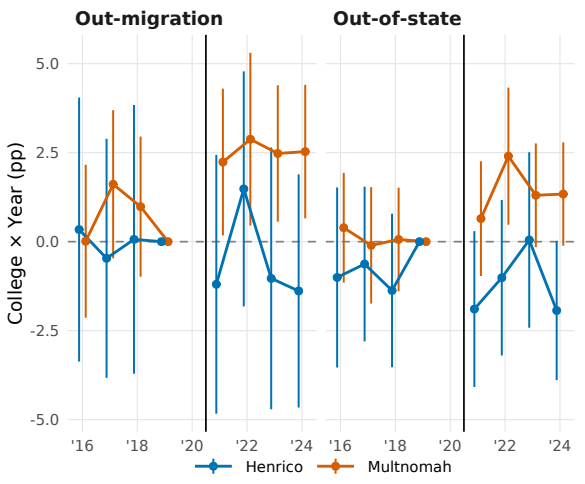






% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Richmond, VA	22.5	20.3	Border county	53	53	46	43
Chesterfield, VA	14.7	16.0	Within state	71	72	47	45
Hanover, VA	9.6	10.4	Bordering state	4	3	27	27
Goochland, VA	3.0	3.3	Top 10 flows	57	57	56	54
Fairfax, VA	1.8	1.5	**In-movers, from:**				
Top origins			Border county	50	49	35	34
Richmond, VA	29.4	28.5	Within state	70	71	39	39
Chesterfield, VA	10.6	10.9	Bordering state	4	4	28	30
Hanover, VA	6.8	6.4	Top 10 flows	54	54	48	48
Fairfax, VA	1.8	2.3					
Goochland, VA	1.7	2.0					

Placebo event study (base 2019, 95% CI)



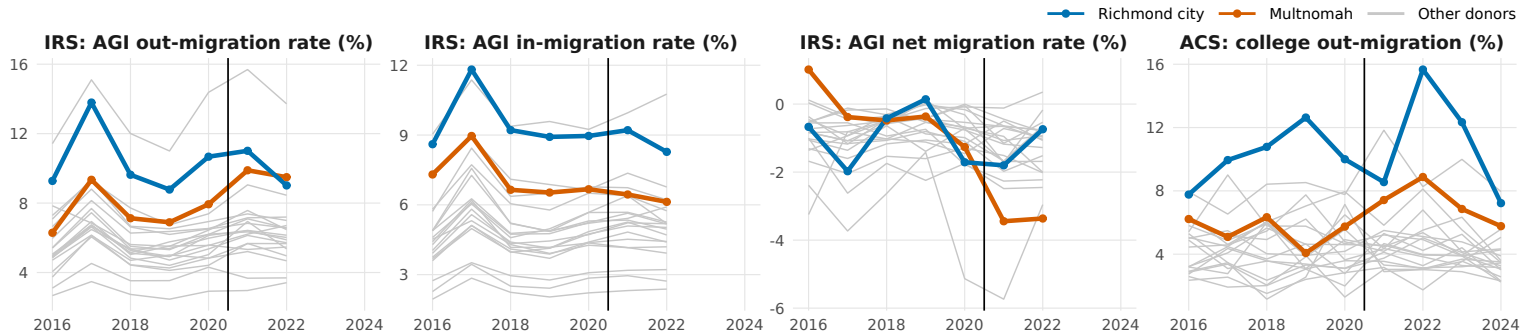
Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.

Richmond city, Virginia

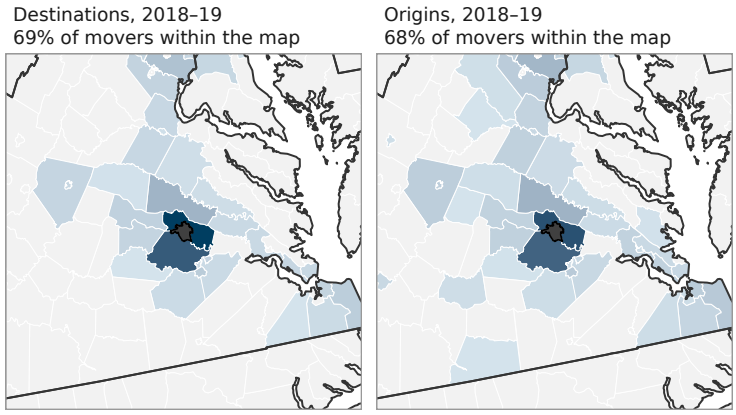
Not used as a placebo: fails the IRS-trend screen (b)

→ = 2018–19 to 2021–22 average	Richmond city	Multnomah	Screens; College × Post DiD	This county	Multnomah
Population, 2020 Census	226,610	815,428	Used as a placebo (both screens)?	NO	
Density (people per sq. mile)	3,782	1,891	Origin county identified in the ACS?	yes	
Urban share of population	100%	99%	(a) Own pre-trend, joint p-value	0.19 — pass	0.39
Median household income (ACS)	$47,250	$69,176	(b) IRS slope distance from Multnomah	0.60 — FAIL	cutoff 0.43
Share under 24 / 65 and over	30% / 13%	27% / 13%	rank among donors (1 = closest)	18 of 20	
State top income-tax rate, 2019 / 2020 (%)	5.75 / 5.75	9.90 / 9.90	AGI out / in slope, 2016–19 (pp/yr)	-0.57 / -0.17	-0.04 / -0.46
Governor's party, 2019 / 2020	D / D	D / D	In NJ or NY (top-rate rises 2020–21)?	no	
Days under stay-at-home order	60	88	Other donor pools	stringency; urban-COVID	
IRS AGI out-migration rate (%)	9.21 → 10.01	7.01 → 9.69	DiD, out-migration: pp (se)	0.54 (1.30)	1.88 (0.57)
IRS AGI in-migration rate (%)	9.07 → 8.74	6.59 → 6.29	DiD, out-of-state: pp (se)	-0.34 (0.64)	1.34 (0.44)
IRS AGI net migration rate (%)	-0.14 → -1.27	-0.42 → -3.40	DiD, in-migration: pp (se)	-0.0000 (0.0010)	-0.0028 (0.0020)

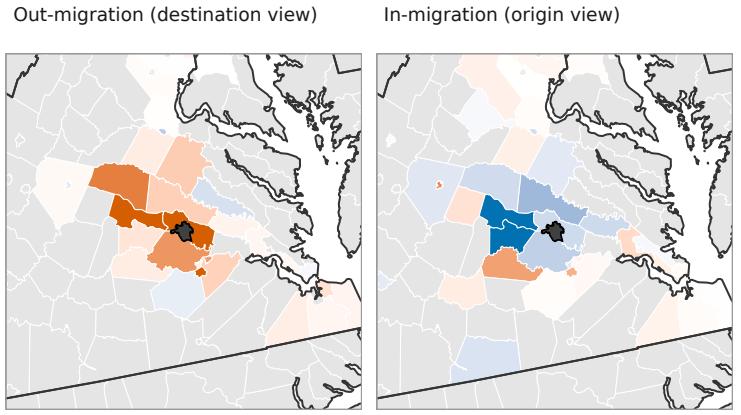
Migration rates before and after 2020



Where its movers went and came from

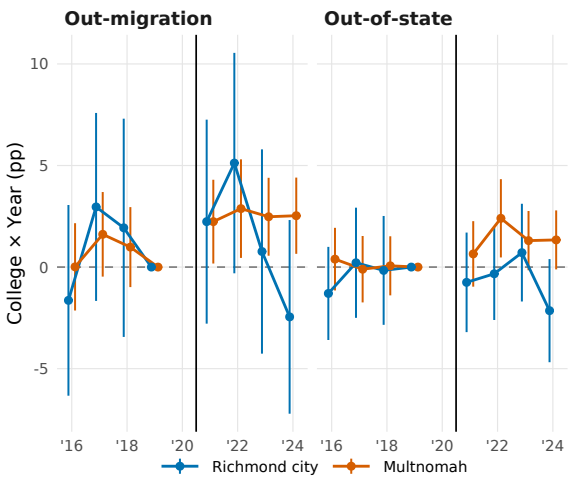


Change in AGI flows, 2018–19 to 2021–22 (as Figure 6)



% of movers	2018–19	2021–22	% of movers	2018–19	2021–22	Mult. pre	Mult. post
Top destinations			**Out-movers, to:**				
Henrico, VA	30.8	28.0	Border county	51	49	46	43
Chesterfield, VA	20.6	21.2	Within state	73	73	47	45
Hanover, VA	2.9	2.9	Bordering state	4	5	27	27
Fairfax, VA	1.6	1.6	Top 10 flows	61	59	56	54
Petersburg, VA	1.0	1.3	**In-movers, from:**				
Top origins			Border county	42	37	35	34
Henrico, VA	23.7	21.1	Within state	71	67	39	39
Chesterfield, VA	18.2	15.6	Bordering state	5	6	28	30
Hanover, VA	2.9	2.4	Top 10 flows	54	50	48	48
Fairfax, VA	2.6	3.2					
District of Columbia, DC	1.2	1.7					

Placebo event study (base 2019, 95% CI)



Sources: IRS SOI county migration files by filing year (rates as % of the county's base-year AGI; mover shares use returns); ACS 1-year PUMS, adults 25+.
Bordering-state shares are lower bounds (IRS disclosure). Change maps: grey = no flow reported in one or both windows.
DiD: 02_did_placebo.do, College × Year with the county as the focal county, 2016–2024 without 2020; bars are 95% confidence intervals.